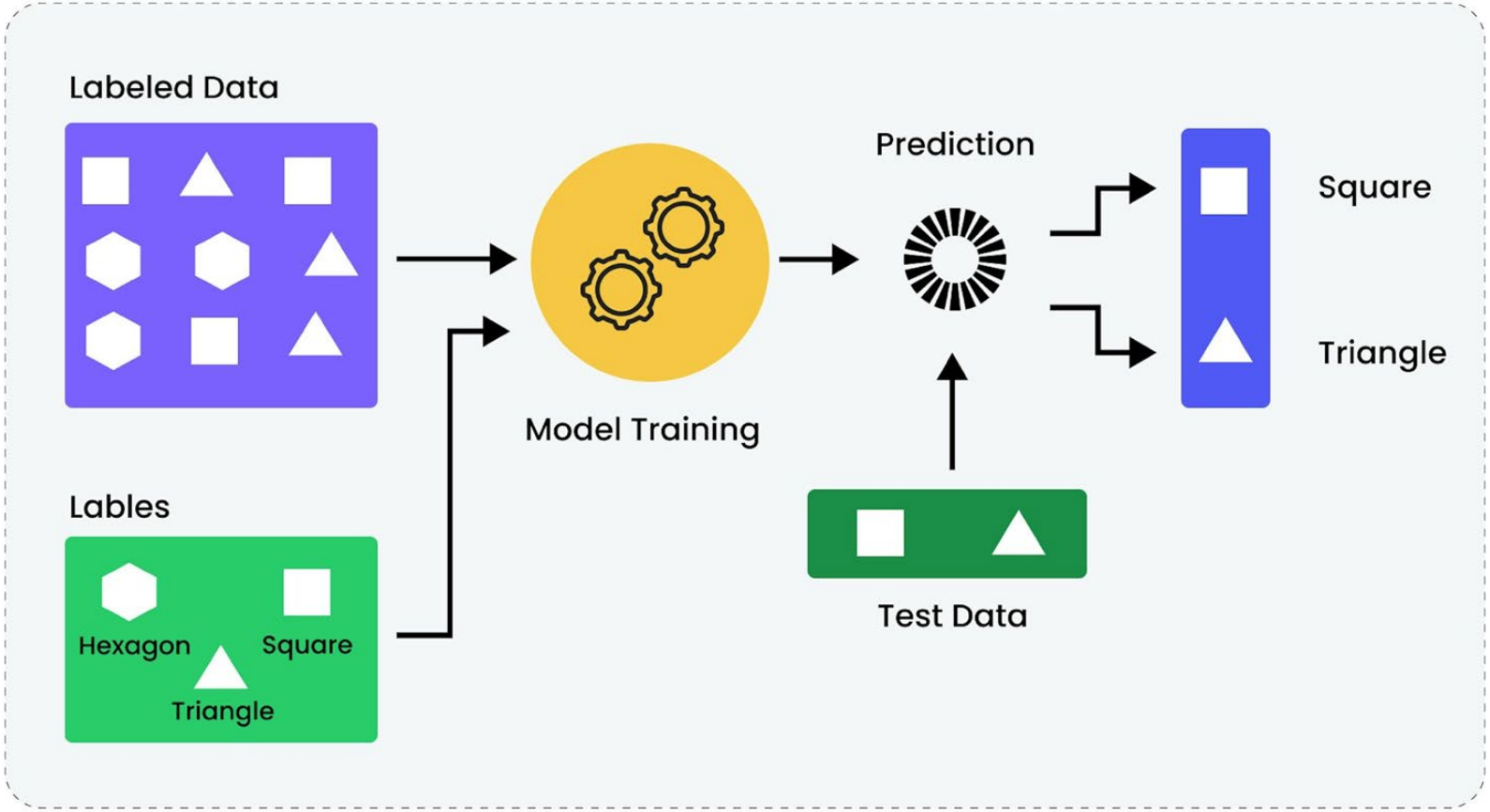
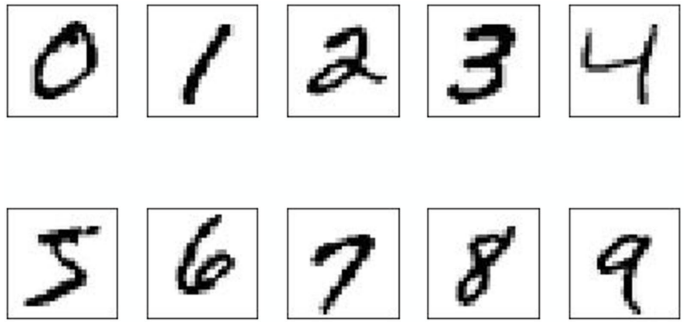
SUPERVISED LEARNING PART 1 : CLASSIFICATION



Supervised Learning

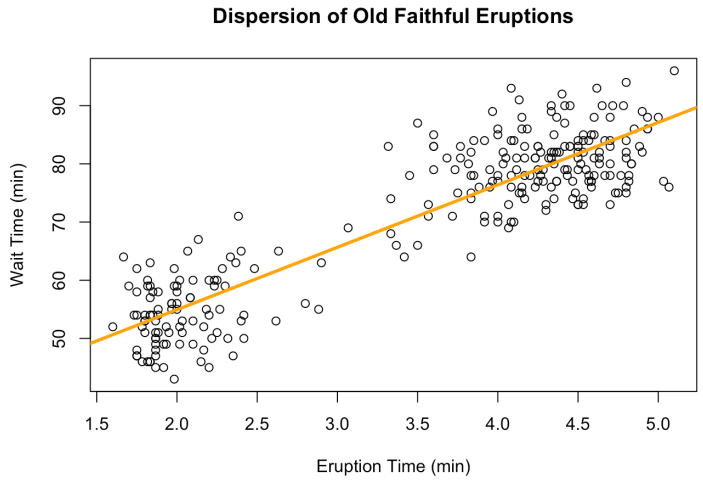
Classification

- Predict categories
- Validation:
  - Accuracy (and many more!)
- Example: Which digit is it?



Regression

- Predict continuous values
- Validation:
  - Mean-squared error (more to come)
- Example: How long must we wait?



K-Nearest Neighbors (KNN)

K-NEAREST NEIGHBORS (KNN)

- A **supervised**, non-parametric method
- Used for **classification and regression**
- Stores the training data (no explicit model)

Main Idea:

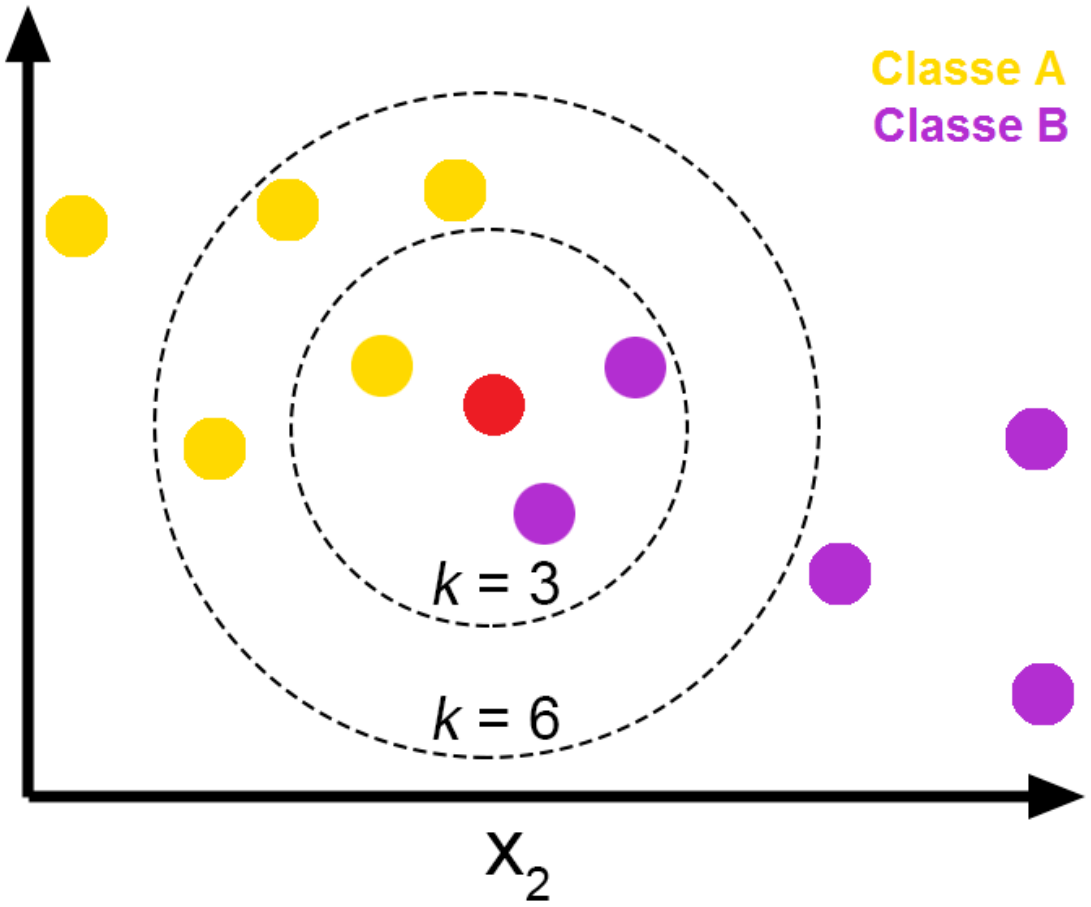
- To predict a new point:
- Find the k closest training points
- Use their labels to make a prediction
- Classification → majority vote

K-NN Algorithm

Steps:

1. Calculate the distance between the new point and all training points.
2. Sort the distances and find the k nearest neighbors.
3. Predict the label based on the neighbors.

- For classification, the majority label among the neighbors is chosen.
- For regression, the average of the neighbors' values is used.



Hyper Parameters

Choosing k :

- Small k : Sensitive to noise, can overfit.
- Large k : Smooths boundaries, but may underfit.

Distance Metrics: Common ones are:

- Euclidean Distance (default in Scikit-learn)
- Manhattan Distance
- Minkowski Distance (generalization)
- Cosine Distance (based on angles)

IMPORTANT Scaling Features: Since KNN is distance-based, ensure features are on a similar scale (e.g., use Min-Max Scaling or Standardization).

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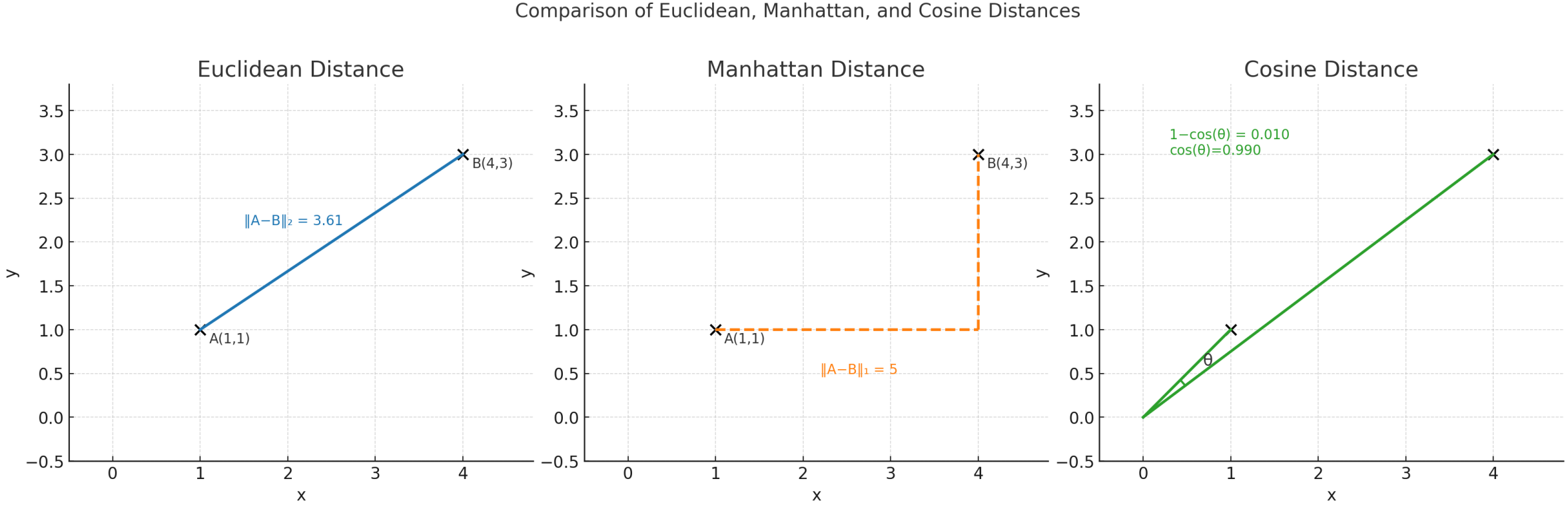
Pros and Cons

Advantages:

- Simple to understand and implement.
- No explicit training phase; training data is directly used.

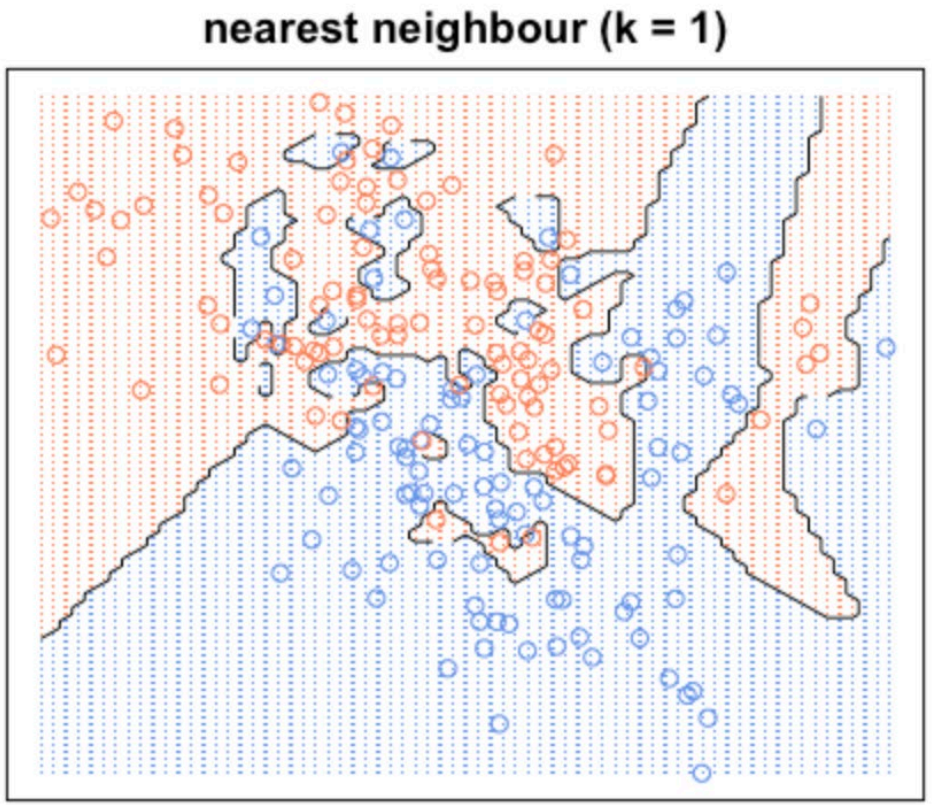
Disadvantages:

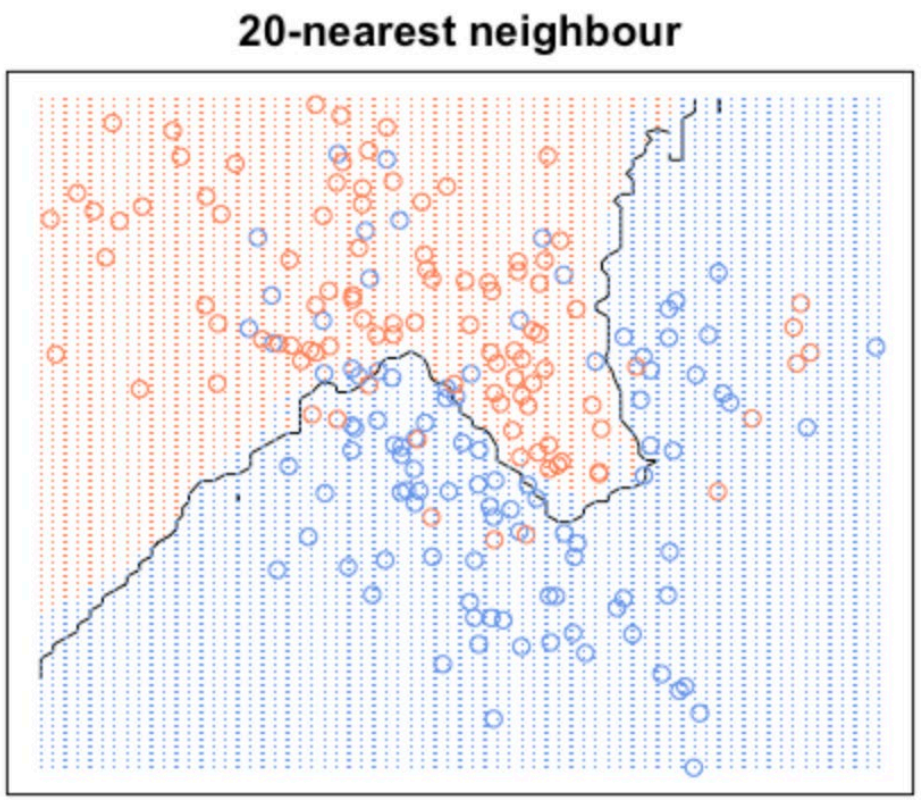
- Computationally expensive for large datasets.
- Sensitive to irrelevant features and feature scaling.



Choice of Distance

- Euclidean most common
- Cosine often used for high-dimensional data





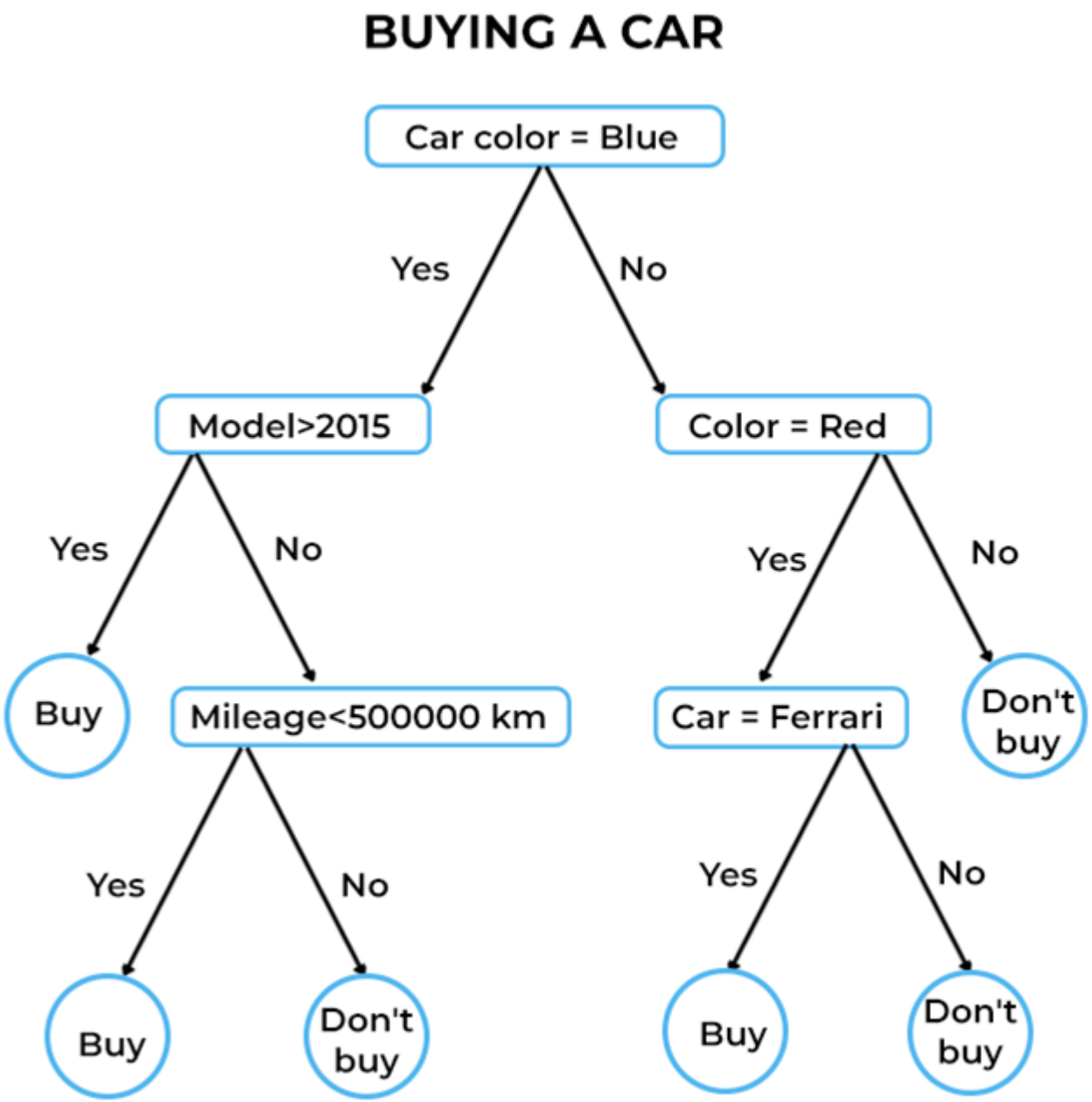
Effect of k on Decision Boundary

https://fderyckel.github.io/machinelearningwithr/knnchapter.html

Decision Trees

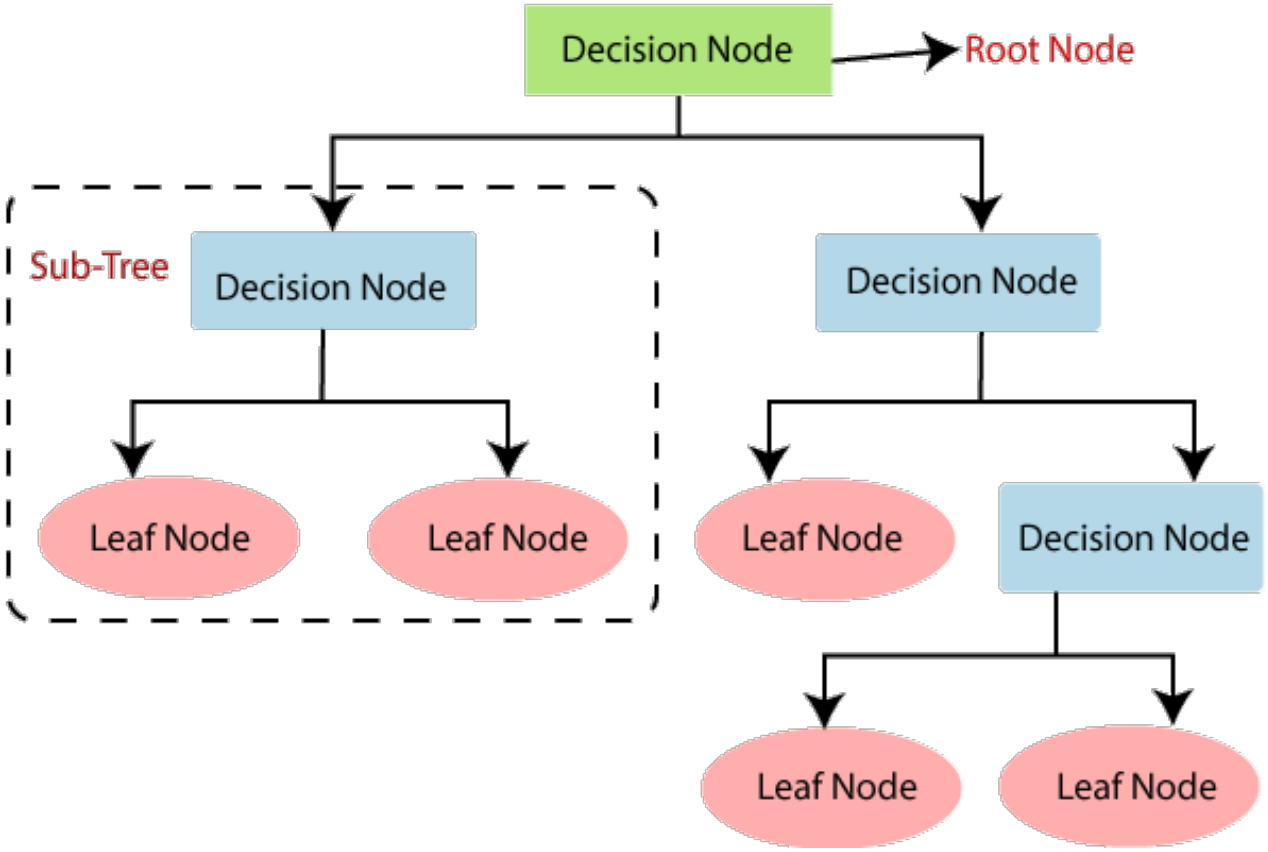
DECISION TREES

- **Classification** and **regression** tasks.
- **They are explainable**
- **Can model non-linear relationships**
- **Handles numeric and categorical data**
- **Works with missing data***
- **No feature scaling required**
- **Prone to data overfitting, instability, bias**



DECISION TREES DEFINITIONS

- **Root Node:** All data starts here.
- **Branch or Sub-Tree:** A subset (partition) of the tree.
- **Splitting**: Dividing a node into two or more sub-nodes.
- **Decision Node:** Created by splitting.
- **Leaf or Terminal Node:** Decisions are mad here.
- **Pruning:** Removing a sub-node from the tree is called pruning.



DECISION TREE HYPERPARAMETERS (Python)

- **max_depth**: The maximum depth of the tree. Limiting this helps prevent overfitting.
- **min_samples_split**: The minimum number of samples required to split an internal node.
- **min_samples_leaf**: The minimum number of samples required to be at a leaf node.
- **max_features**: The number of features to consider when looking for the best split.

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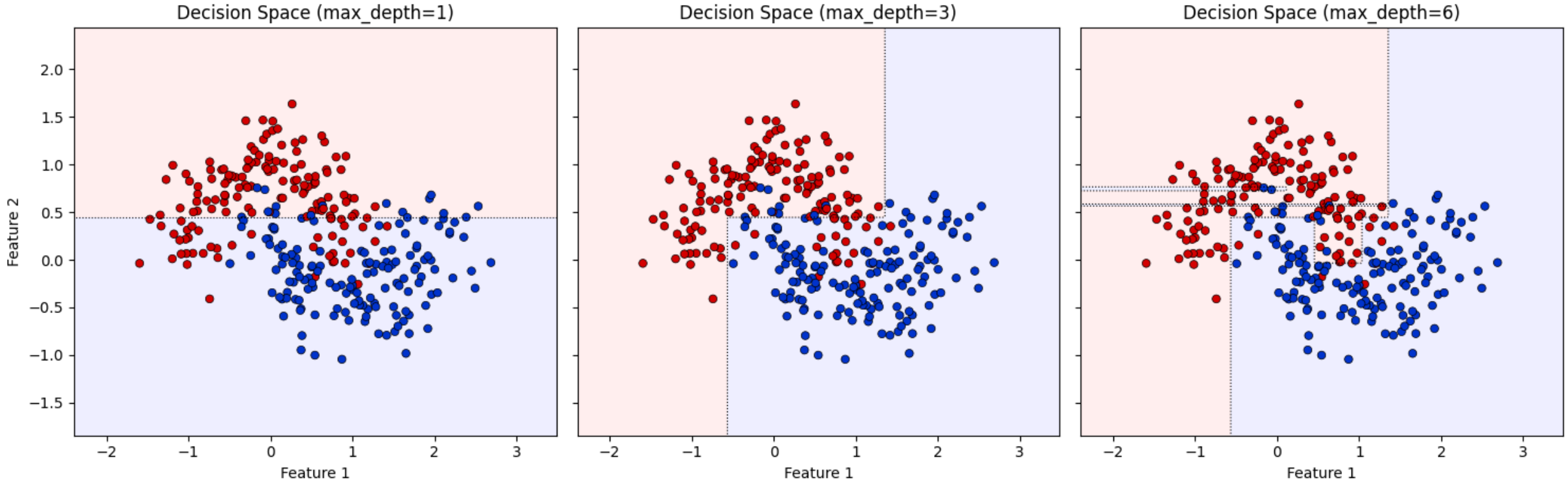
Decision Tree Decision Boundary by Max Depth



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TREE BUILDING PROCESS

1. **Start at the root**: Evaluate all possible splits based on features.

2. **Choose the best split** based on a criterion (How to choose best?).

3. **Repeat** the process recursively until a stopping criterion is met (e.g., max depth, minimum samples per leaf).

Splitting Criteria:

1. **Gini Impurity** (used for classification):

$$Gini = 1 - \sum_{i=1}^{C} p_i^2$$

Where p_i is the proportion of class i in the node.

2. **Entropy** (used for classification):

$$Entropy = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

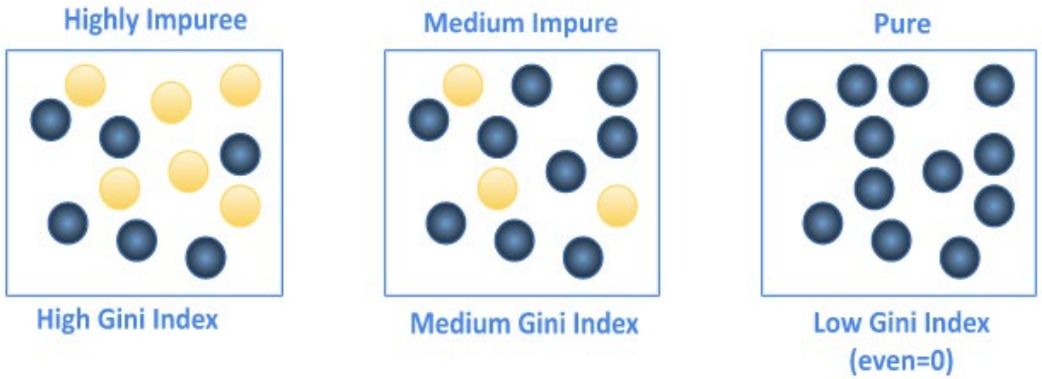
3. **Mean Squared Error (MSE)** (used for regression):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where y_i is the true value and $\hat{y}_i$ is the predicted value.

GINI IMPURITY

- The **Gini Impurity** measure quantifies the "impurity" or **impurity score** of a node.
- A **lower Gini score** means a **purer** node.



Formula:

For a set of classes, the Gini Impurity is calculated as:

$$Gini(D) = 1 - \sum_{i=1}^{C} p_i^2$$

Where:

- D is a dataset.
- C is the number of unique classes.
- p_i is the probability (or proportion) of class i in the dataset D.

- If all data points in a node belong to a single class, the Gini Impurity is 0 (perfect purity).
- If the data points are evenly distributed among all classes, the Gini Impurity is at its maximum value (1 – 1/C)

Example Calculation:

Let's say we have a dataset of 10 samples belonging to two classes: Class 1 (6 samples) and Class 2 (4 samples)

1. **Proportions**:
   - $p_1 = \frac{6}{10} = 0.6$ (Class 1)
   - $p_2 = \frac{4}{10} = 0.4$ (Class 2)
2. **Gini Impurity**:

$$Gini(D) = 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 1 - 0.52 = 0.48$$

ENTROPY

- **Entropy** is another measure of impurity or disorder, derived from information theory. It is used in decision trees to decide the best split by evaluating how much uncertainty (or entropy) is present in the data.

The **Entropy** of a dataset is defined as:

$$Entropy(D) = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

Where:

- D is a dataset.
- C is the number of unique classes.
- p_i is the proportion of class i in dataset D.

How to Interpret Entropy:

- If all samples in a node belong to the same class, entropy is 0 (pure node).
- If the samples are evenly distributed among all classes, entropy reaches its maximum, which is $\log_2(C)$ (where C is the number of classes).

Example Calculation:

Let's consider the same dataset of 10 samples with 6 samples of Class 1 and 4 samples of Class 2.

$$Entropy(D) = -(0.6 \log_2(0.6) + 0.4 \log_2(0.4))$$

Using a calculator:

$$Entropy(D) = -(0.6 \times -0.737 + 0.4 \times -1.322) = 0.442 + 0.528 = 0.970$$

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Entropy vs. Gini Impurity

- Gini Impurity
  - Faster to compute (no logarithms).
  - Prefers pure splits, leading to shallower trees.
  - Default in CART (e.g., Scikit-learn).

- Entropy
  - Slower (uses logarithms) but handles rare classes better.
  - Encourages more balanced splits, sometimes leading to deeper trees.
  - Used in ID3, C4.5 (Information Gain-based methods).

- Key Takeaways:
  - Use Gini when speed and simplicity matter.
  - Use Entropy for imbalanced datasets and finer splits.
  - Differences are usually small in practice.

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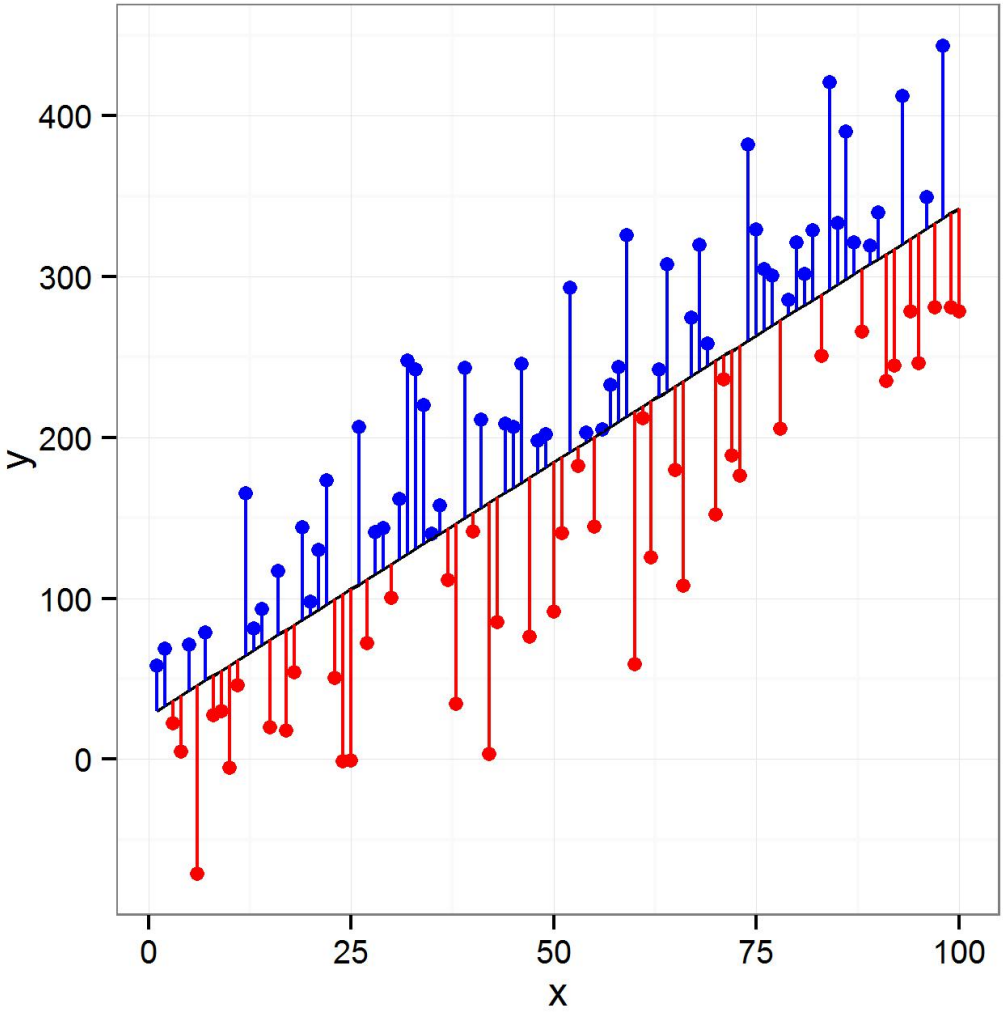
Linear Regression

LINEAR REGRESSION



Linear Regression

- One of the **simplest and most widely used** machine learning models
- Used for **regression problems** (predicting numeric values)
- Learns a relationship between:
  - **Input features** X
  - **Continuous target variable** y
- Examples:
  - Predict house price from size
  - Predict temperature tomorrow

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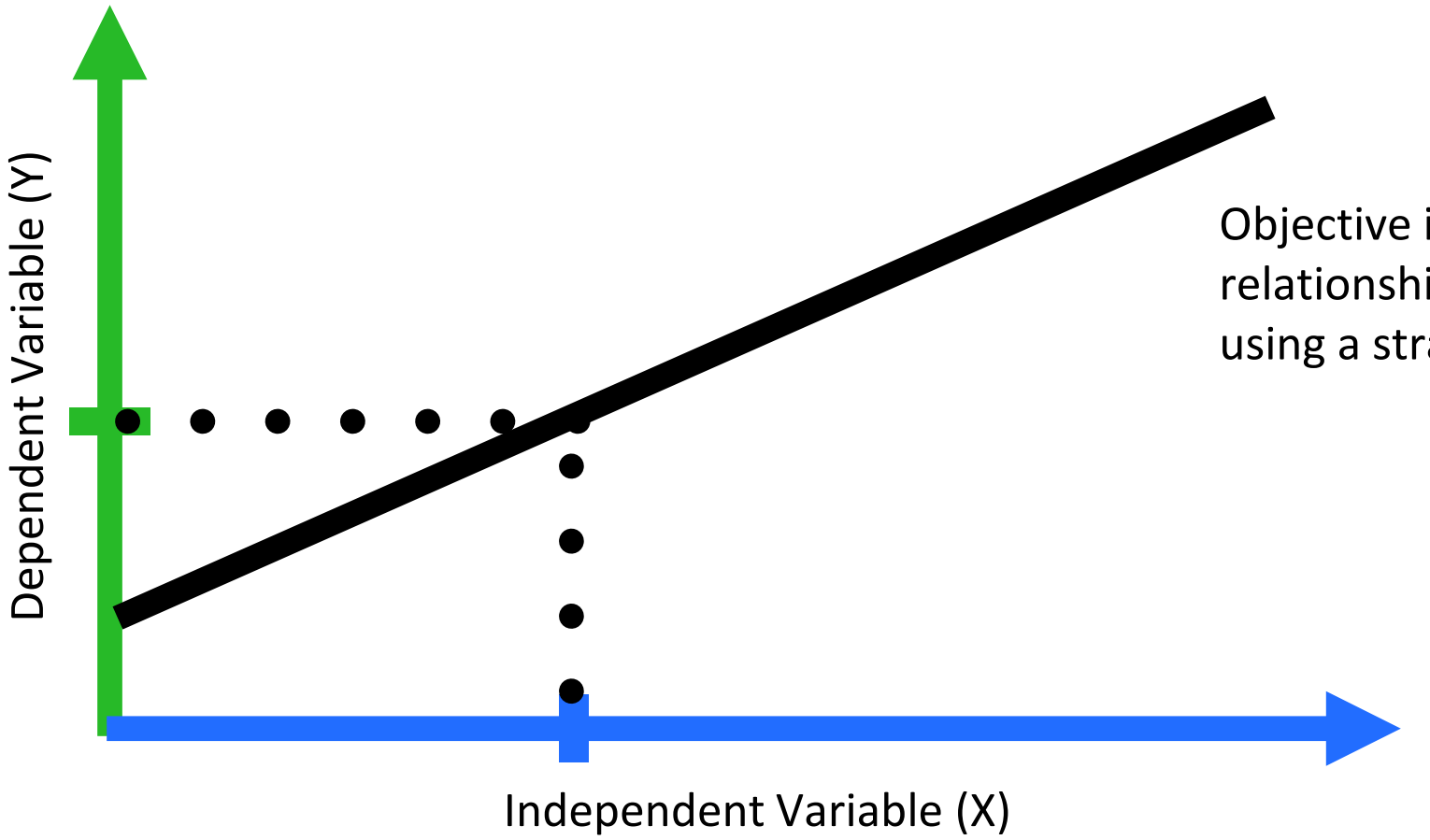
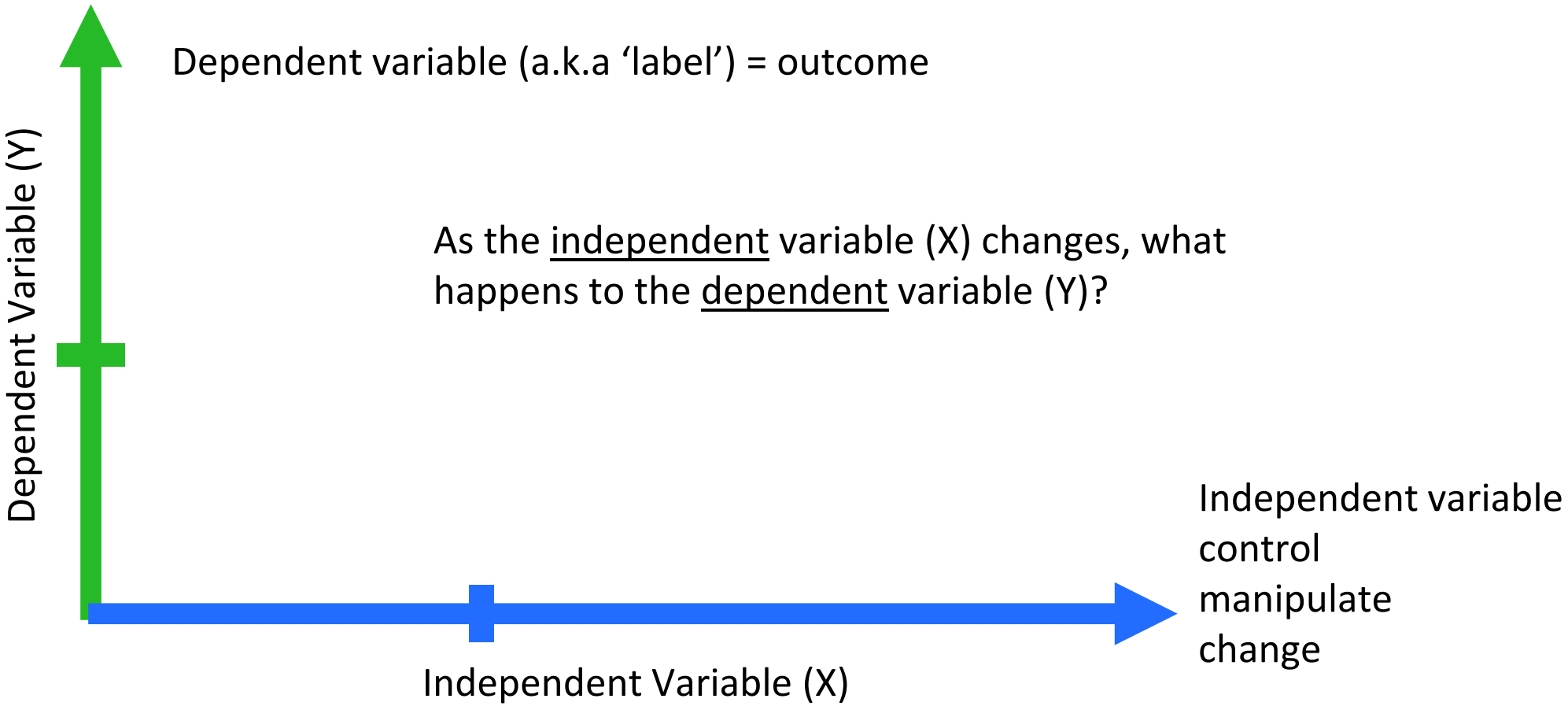
INTRODUCTION TO LINEAR REGRESSION

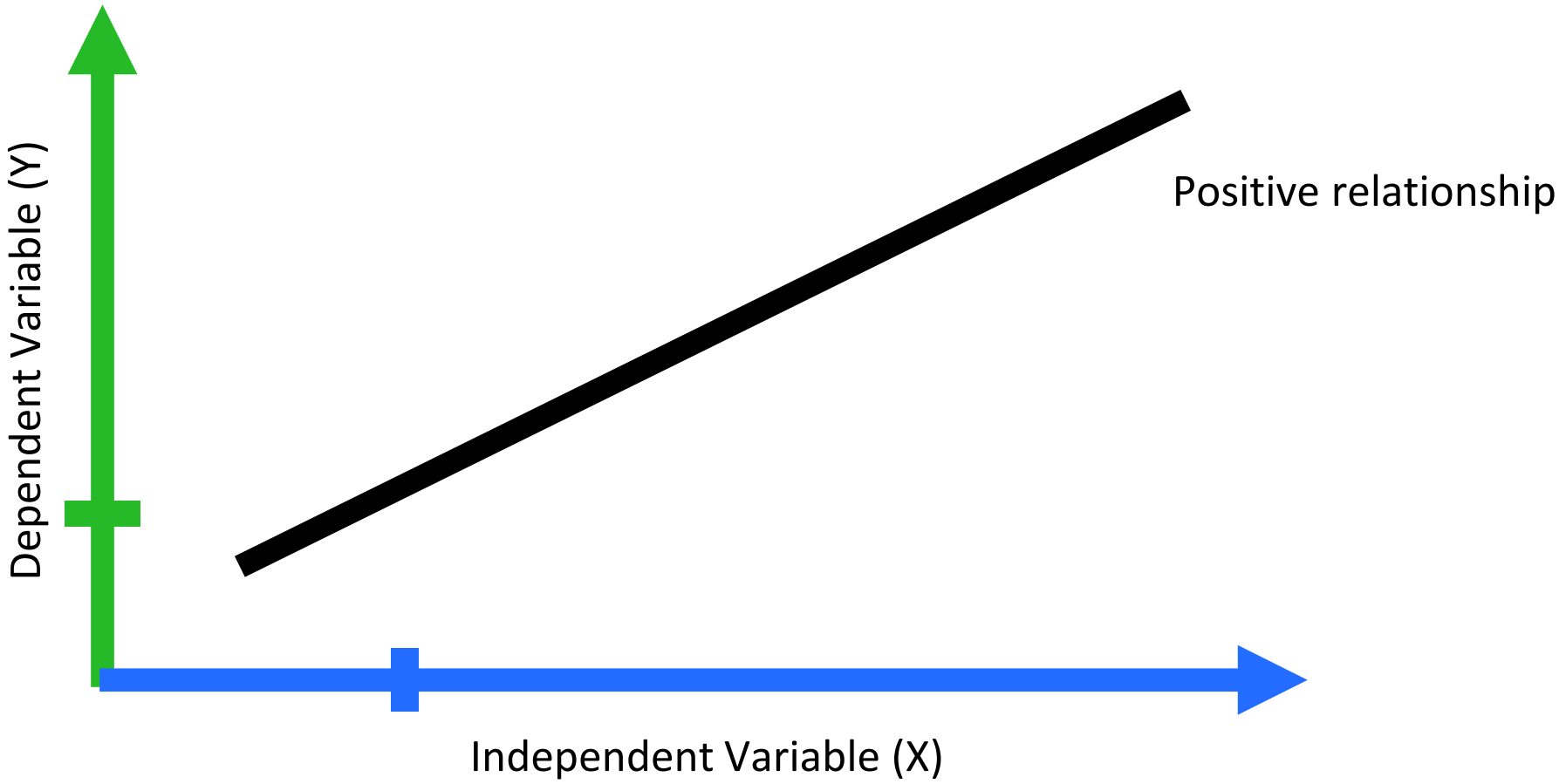


Objective is to describe a relationship between X & Y using a straight line

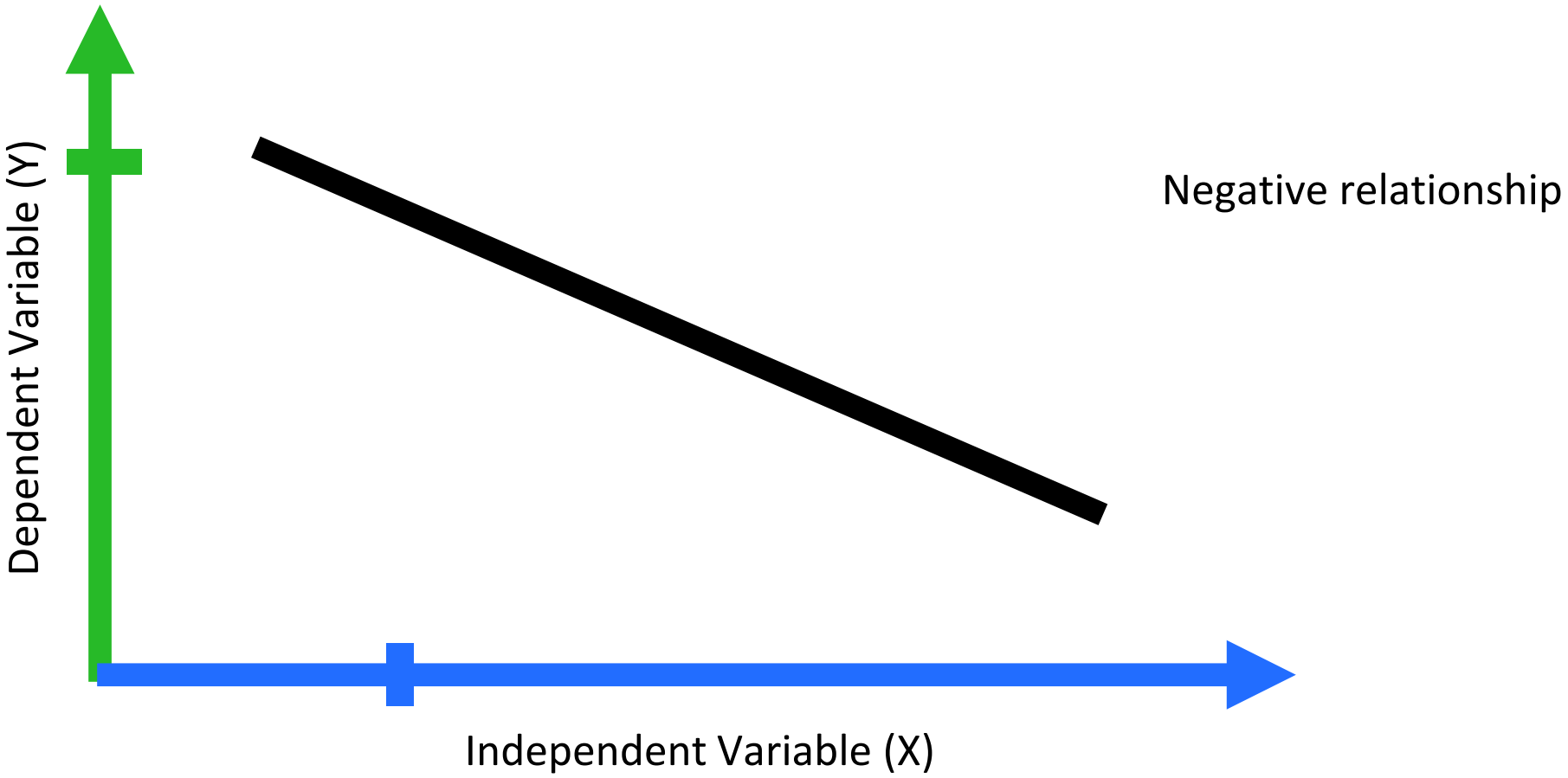
INTRODUCTION TO LINEAR REGRESSION



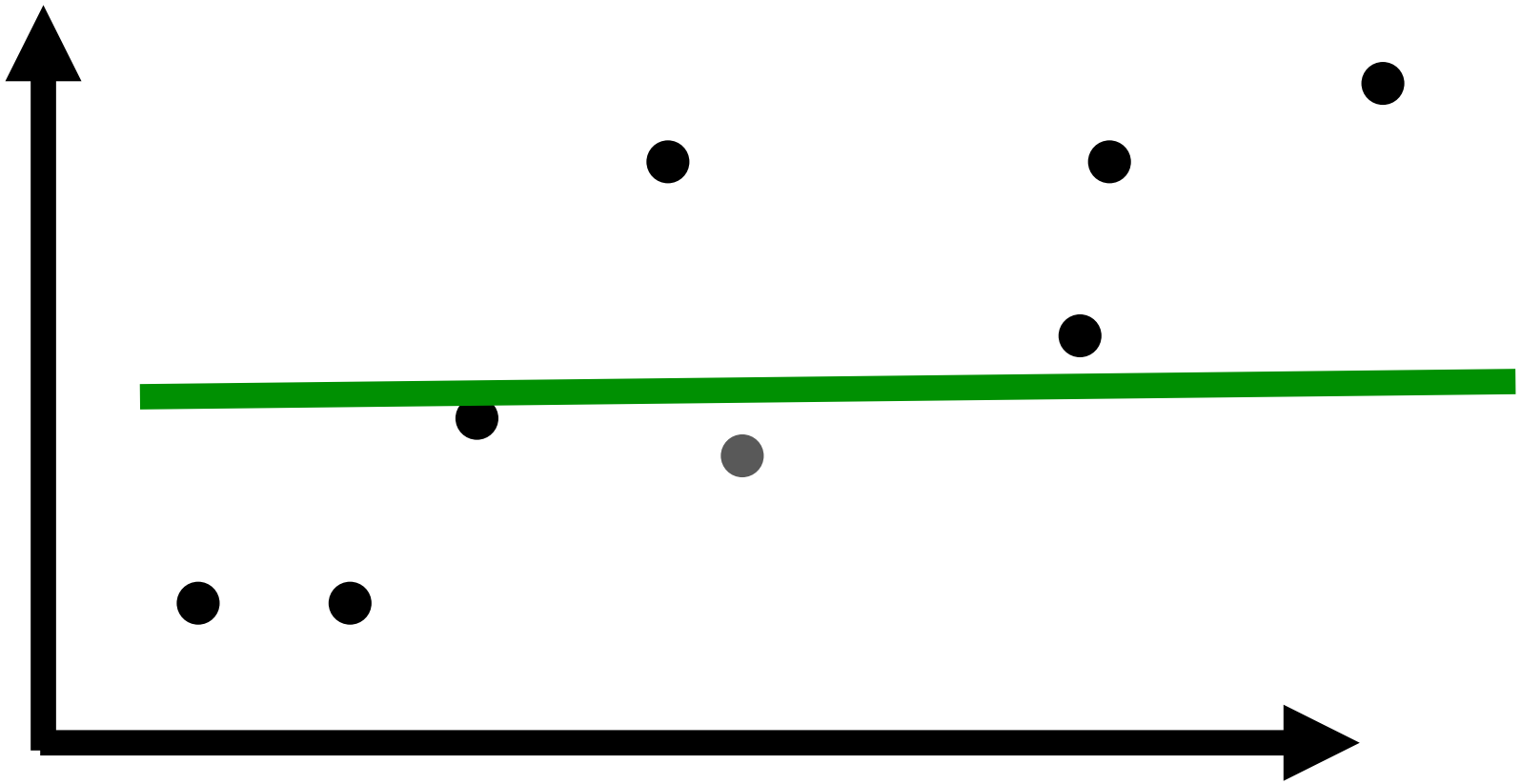
INTRODUCTION TO LINEAR REGRESSION



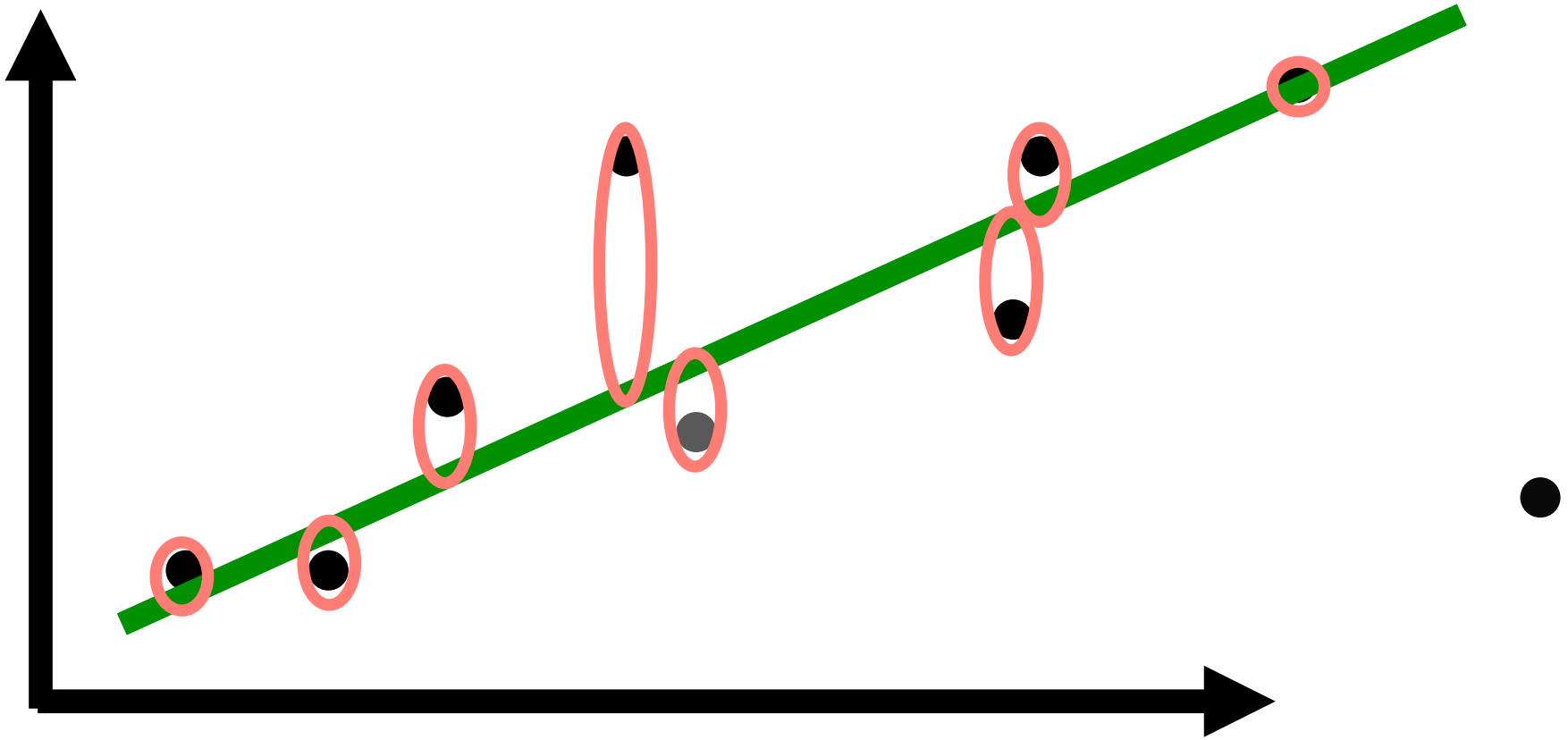
INTRODUCTION TO LINEAR REGRESSION



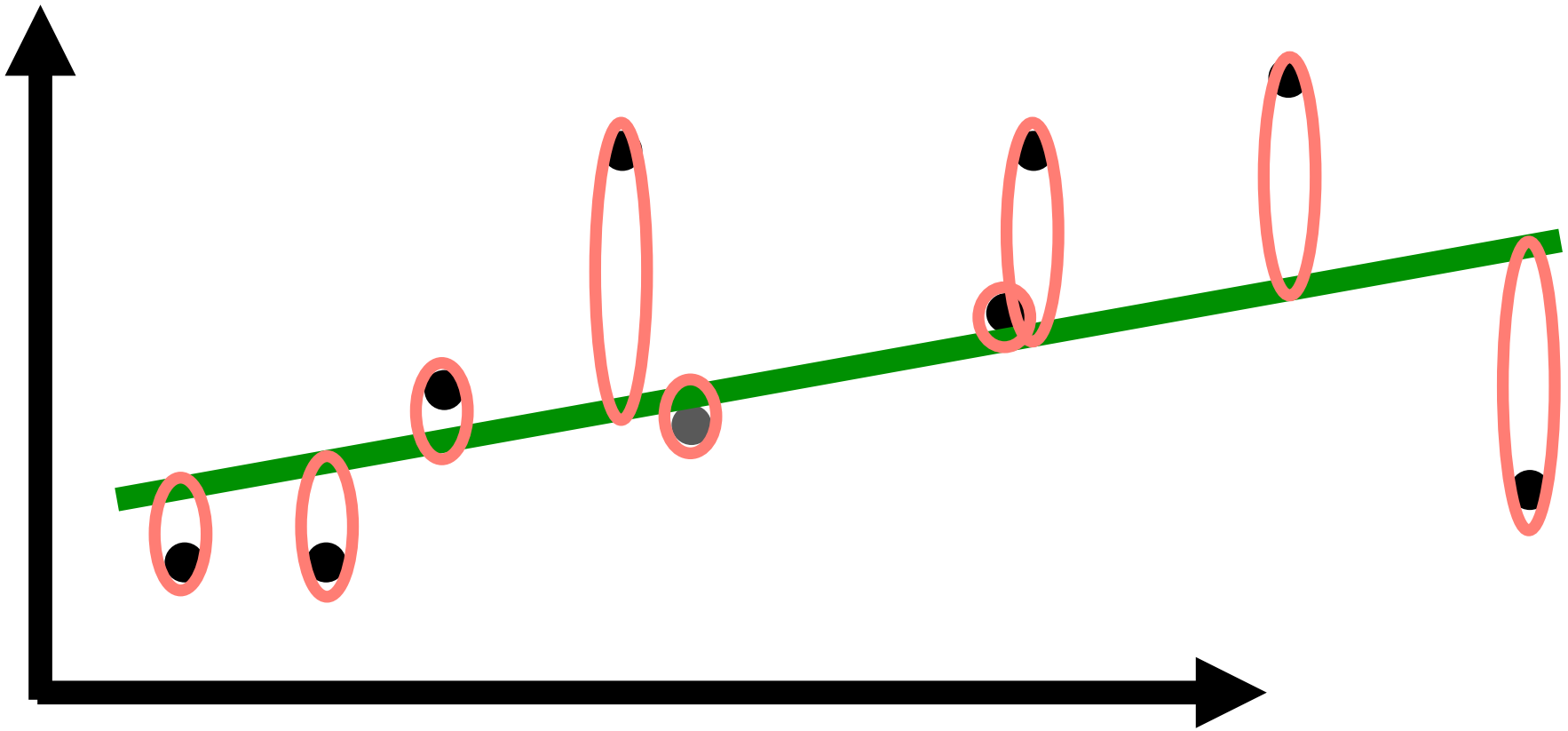
THE RUBBERBAND BOARD



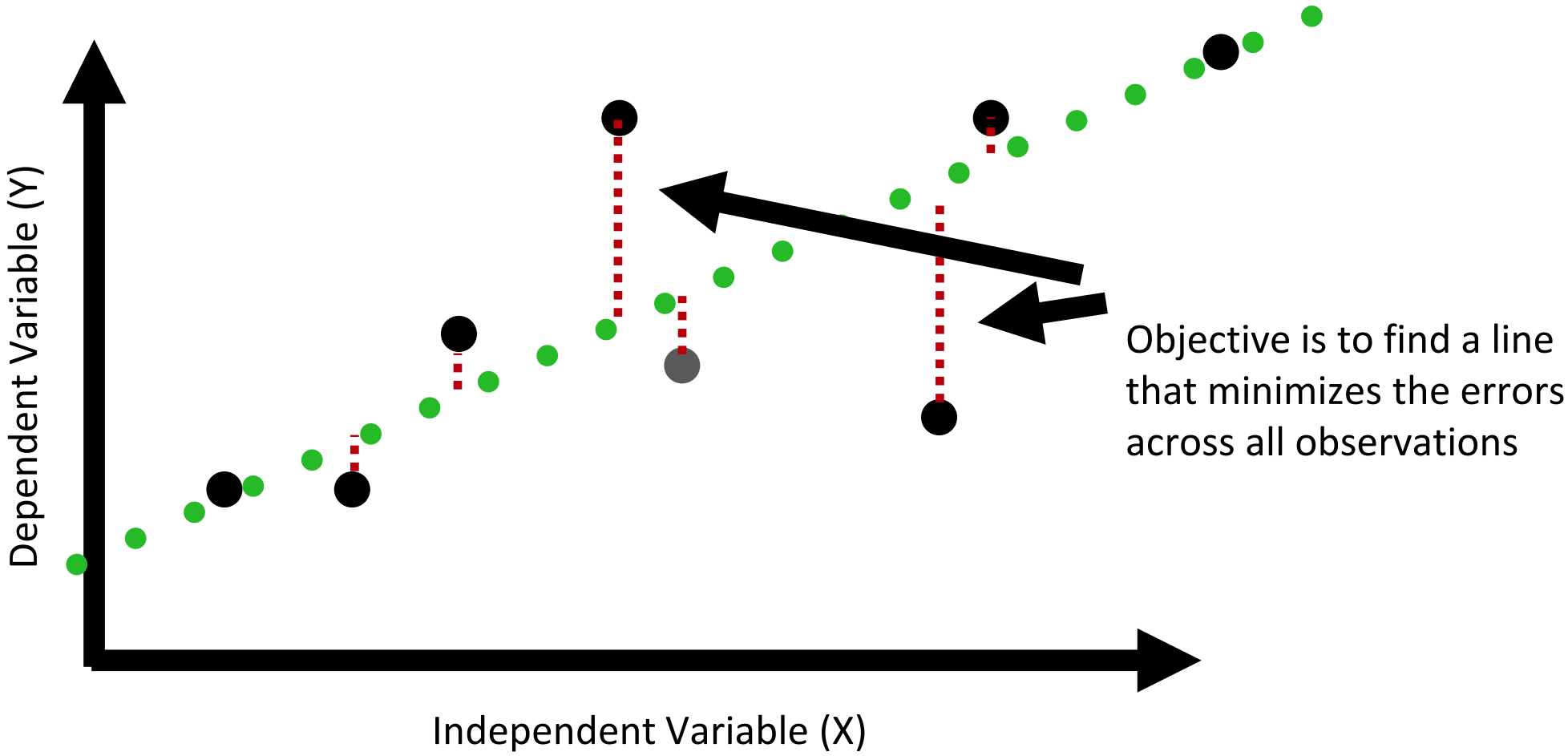
THE RUBBERBAND BOARD



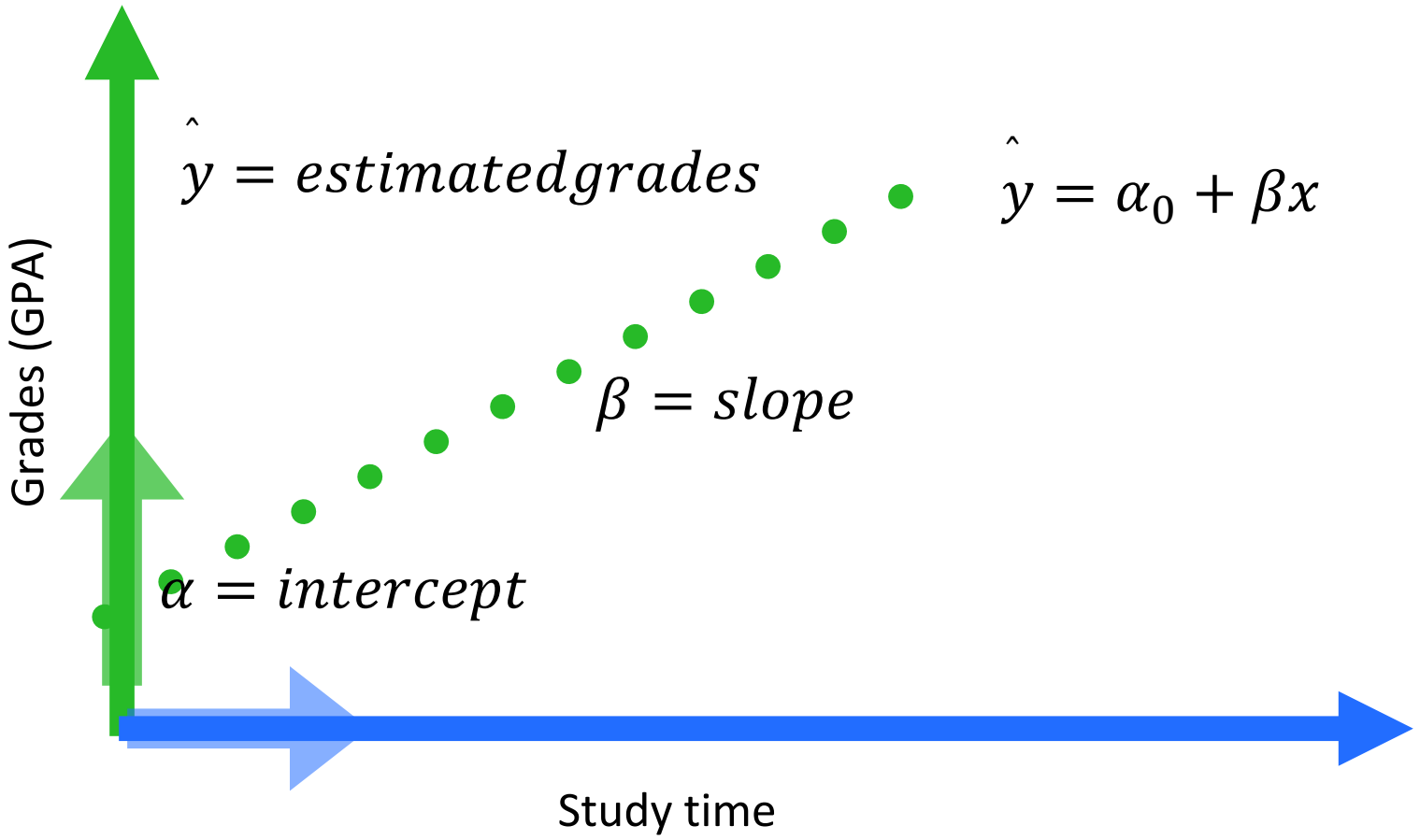
THE RUBBERBAND BOARD



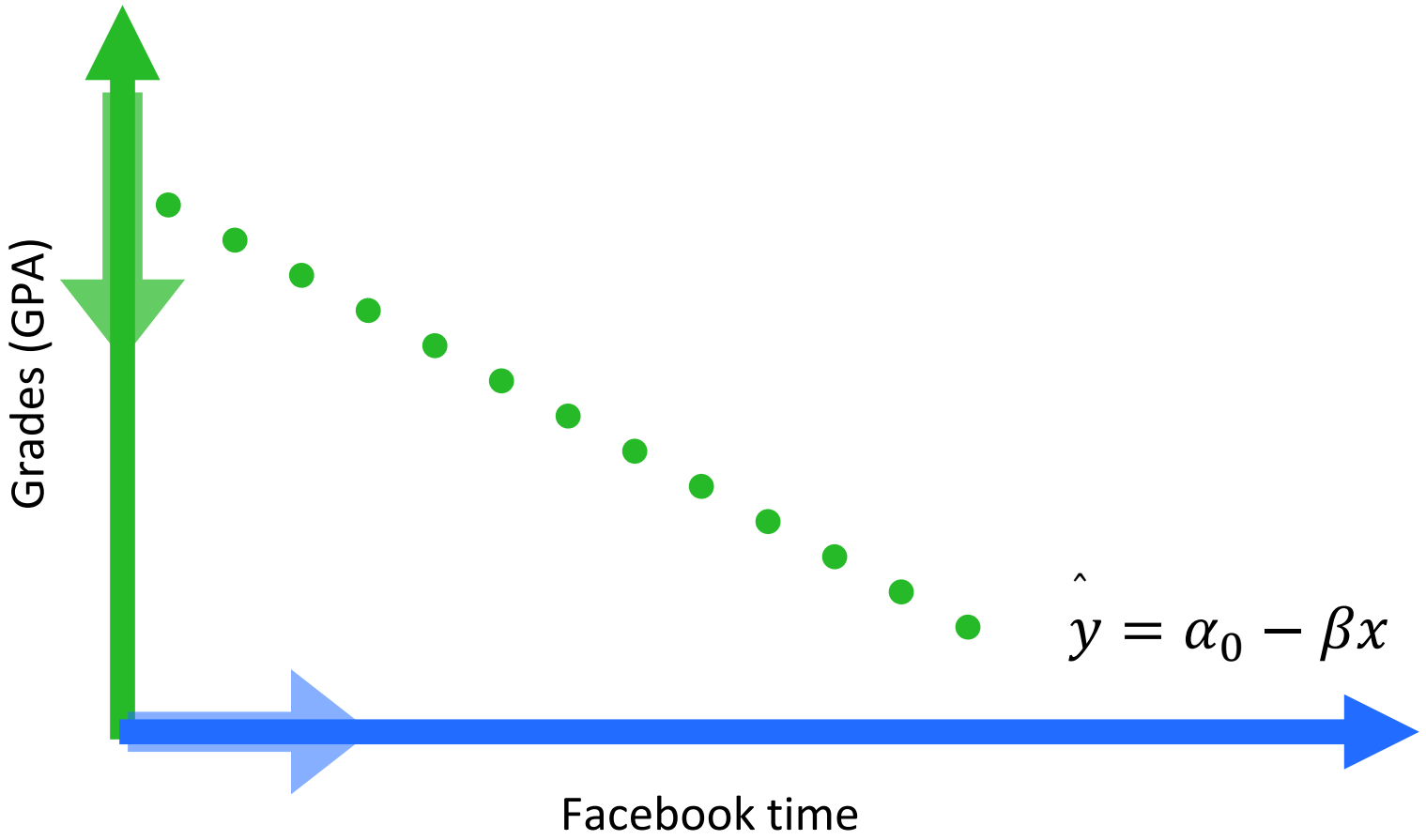
PUTTING IT ALL TOGETHER



EXAMPLE: STUDY TIME



EXAMPLE: STUDY TIME



THE GOOD

- Good for predicting numerical labels or outcomes
- Good for linear relationships and extrapolation
- Can be adapted for curved lines (e.g., Log) for special purposes

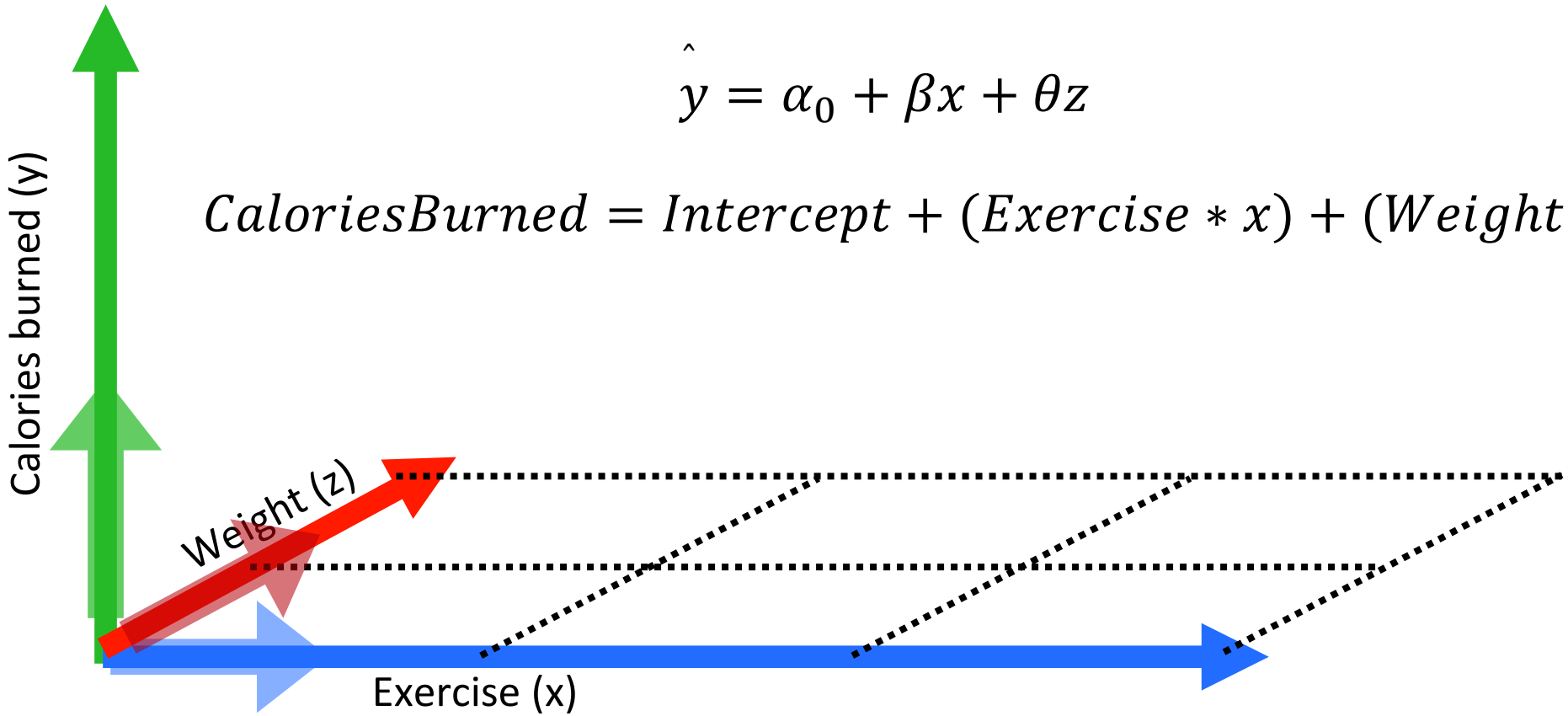
THE BAD

- Very finicky about data.
  - All attributes must be numerical (or converted)
  - No missing values
- Very large outliers can easily ruin a model because errors are squared
- Highly correlated attributes can be problematic
- Attributes with substantially different scales introduce issues

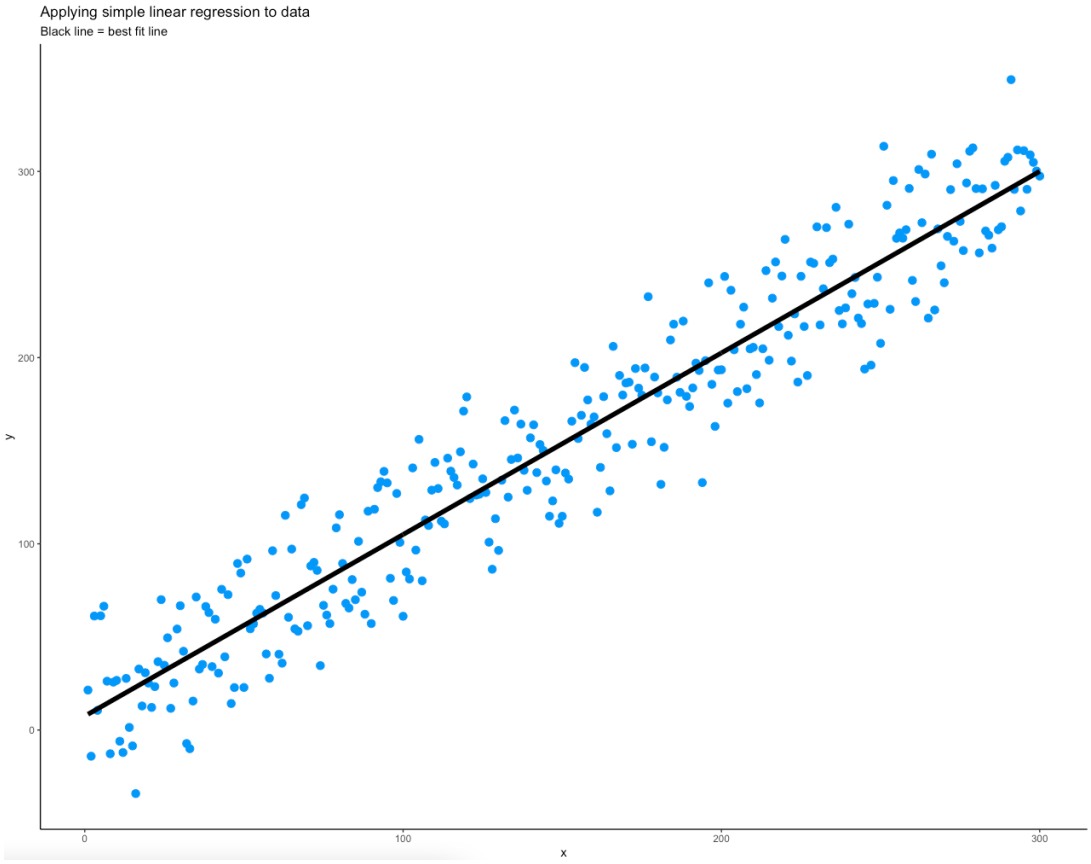
THERE CAN BE MORE THAN ONE INDEPENDENT VARIABLE

$$\hat{y} = \alpha_0 + \beta x + \theta z$$

$$CaloriesBurned = Intercept + (Exercise * x) + (Weight * z)$$

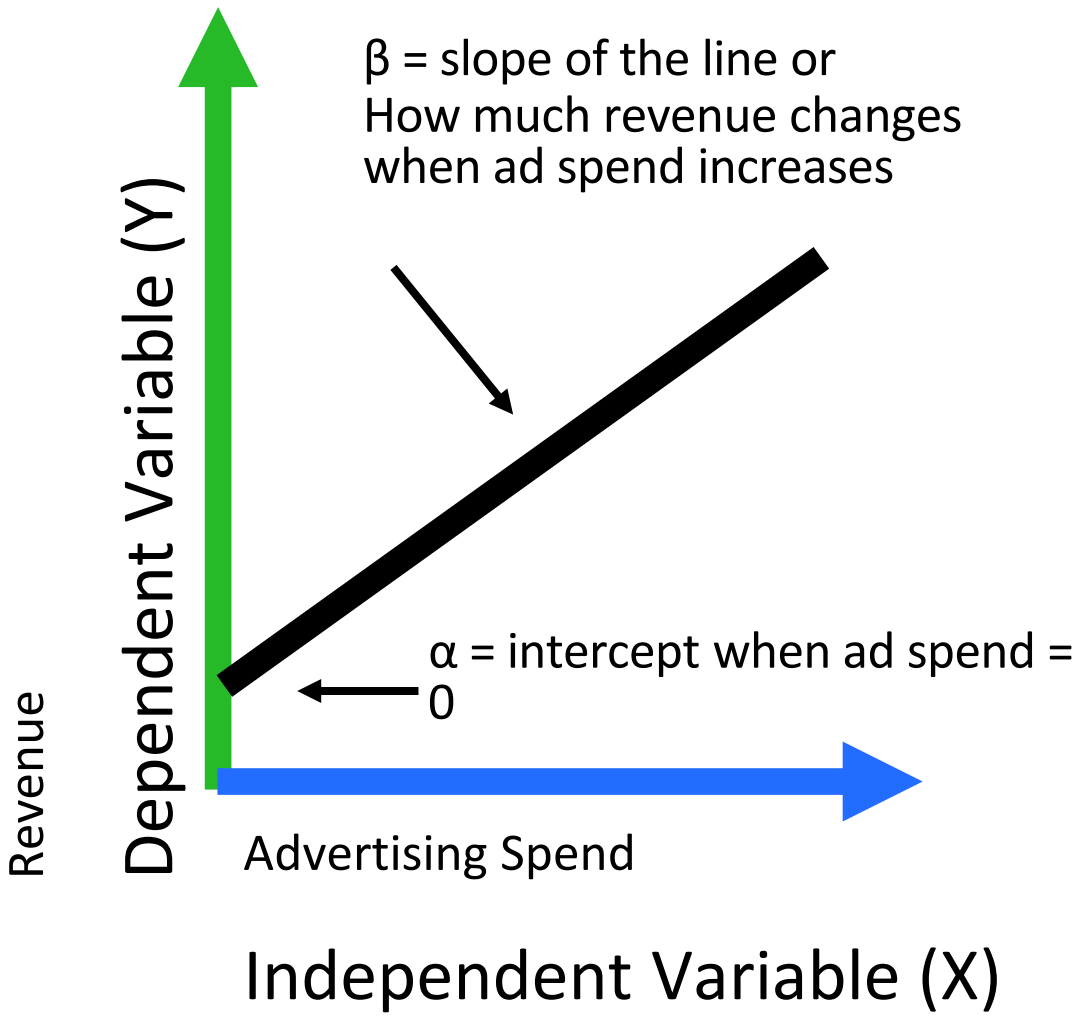


REAL LIFE EXAMPLES



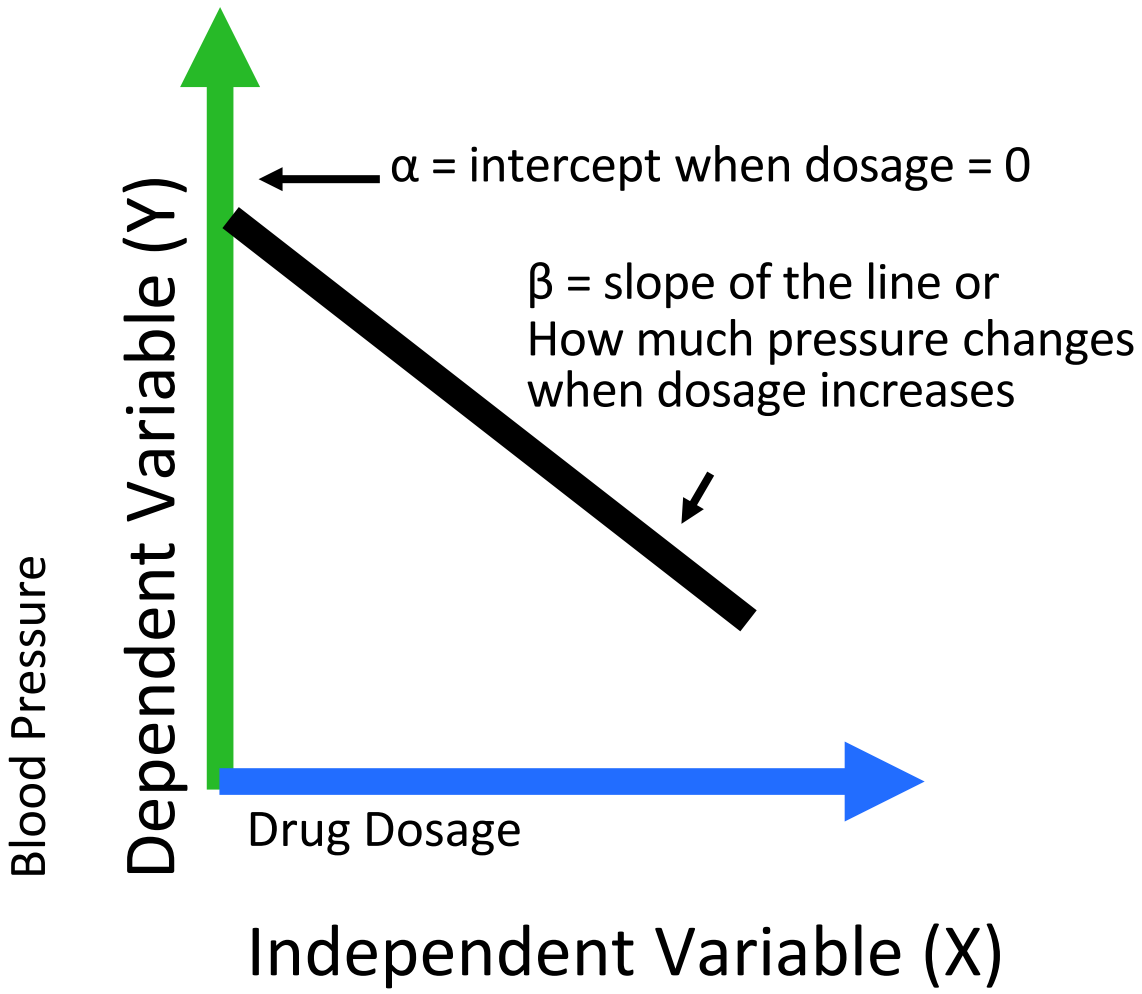
#1 - THE RELATIONSHIP BETWEEN ADVERTISING & REVENUE

- Businesses often use linear regression to understand the relationship between advertising spending and revenue.
- revenue = α + β(ad spending)
- The coefficient α (intercept) would represent total expected revenue when ad spending is zero.
- The coefficient β (slope) would represent the average change in total revenue when ad spending is increased by one unit (e.g. one dollar).
- If β is negative, it would mean that more ad spending is associated with less revenue.
- If β is close to zero, it would mean that ad spending has little effect on revenue.
- And if β is positive, it would mean more ad spending is associated with more revenue.



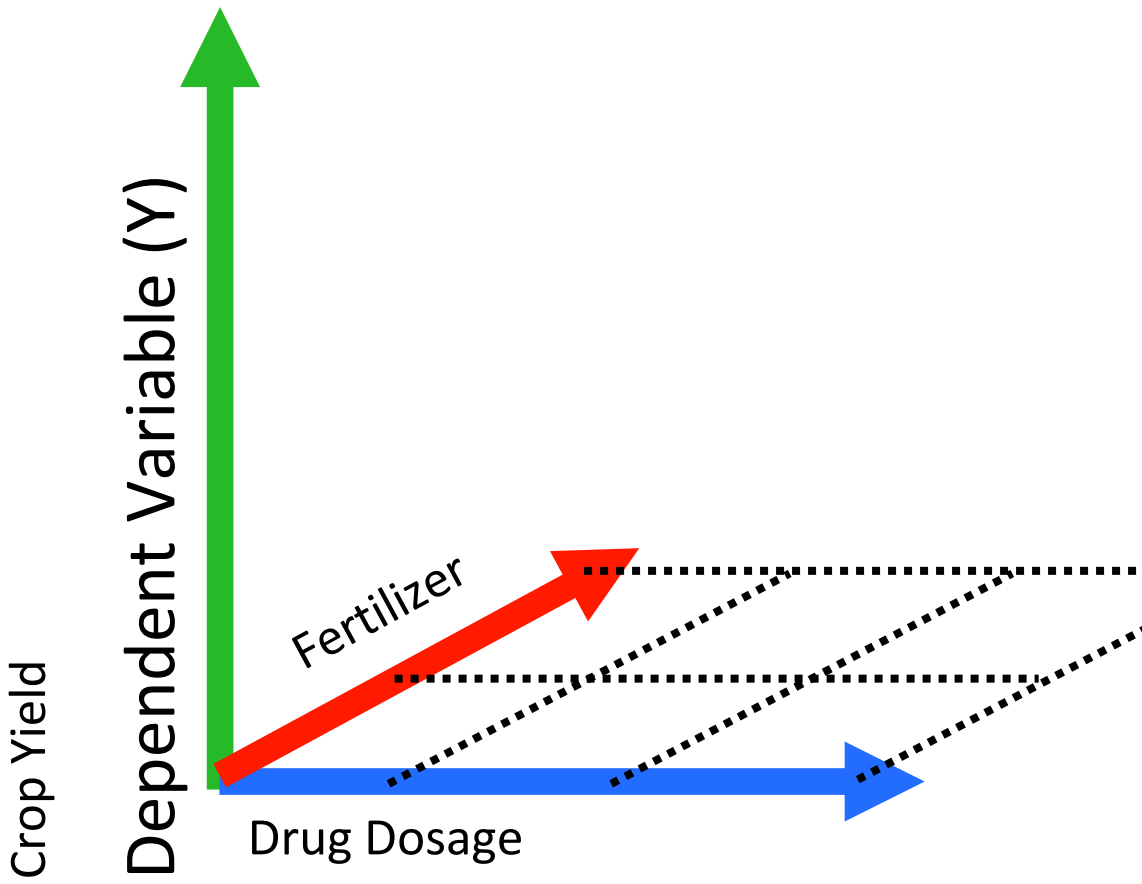
#2 - THE EFFECTS OF DRUG DOSAGE

- Medical researchers often use linear regression to understand the relationship between drug dosage and blood pressure of patients.
- blood pressure = $\alpha + \beta$(dosage)
- The coefficient α would represent the expected blood pressure when dosage is zero.
- The coefficient β would represent the average change in blood pressure when dosage is increased by one unit.
- If β is negative, it would mean that an increase in dosage is associated with a decrease in blood pressure.
- If β is close to zero, it would mean that an increase in dosage is associated with no change in blood pressure.
- If β is positive, it would mean that an increase in dosage is associated with an increase in blood pressure.



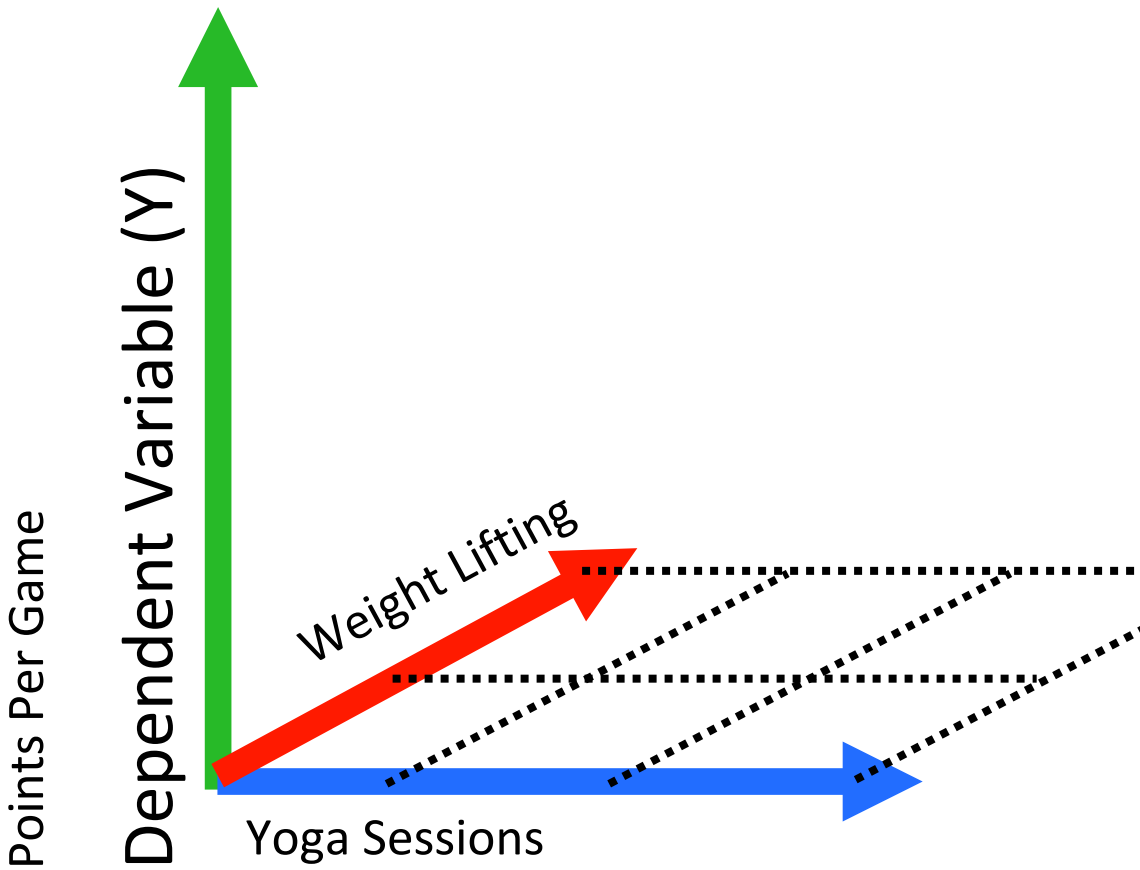
#3 - THE EFFECTS OF FERTILIZER AND WATER ON CROP YIELD

- Agricultural scientists often use linear regression to measure the effect of fertilizer and water on crop yields.
- For example, scientists might use different amounts of fertilizer and water on different fields and see how it affects crop yield.
- crop yield = $\alpha + \beta 1$(amount of water) + $\beta 2$(amount of fertilizer)
- The coefficient α would represent the expected crop yield with no fertilizer or water.
- The coefficient $\beta 1$ = average change in crop yield when fertilizer is increased by one unit
- The coefficient $\beta 2$ would represent the average change in crop yield when water is increased by one unit



#4 - THE EFFECTS OF DIFFERENT TRAINING REGIMENS ON PLAYER PERFORMANCE

- Data scientists in the NBA might analyze how different amounts of weekly yoga sessions and weightlifting sessions affect the number of points a player scores. points scored = α + β1(yoga sessions) + β2(weightlifting sessions)
- The coefficient α = the expected points scored for a player who participates in zero yoga sessions and zero weightlifting sessions.
- The coefficient β1 would represent the average change in points scored when weekly yoga sessions is increased by one.
- The coefficient β2 would represent the average change in points scored when weekly weightlifting sessions is increased by one.



Multiple Linear Regression

In practice, we usually have **multiple features**.

Example:

- Predict house price using:
  - size
  - number of bedrooms
  - location score
- Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$$

Each feature gets a **weight (coefficient)**.

Logistic Regression

LOGISTIC REGRESSION MATH BACKGROUND

- Problem: Predict binary outcome $y \in \{0,1\}$
- Why not linear regression?
- ✗ Unbounded predictions: $\hat{y} \in (-\infty, \infty)$, not $[0,1]$
- ✗ Inappropriate error model: assumes Normal errors

- BUT, it provides a useful starting point with a linear combination of features

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LOGISTIC REGRESSION MATH BACKGROUND

- **Odds and Log-Odds:**
- Odds represent the ratio of the probability of an event occurring to it not occurring:

$$\text{Odds} = \frac{p}{1-p}$$

- Taking the logarithm of the odds yields the log-odds (logit function):

$$\text{Logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$$

- This linear relationship between log-odds and input features forms the basis of logistic regression.

- Logistic Regression models the **probability** that an input belongs to a particular class.
- The linear model outputs real-valued numbers.
- Predicted values are mapped between 0 and 1.
- The logistic (sigmoid) function is defined as:

$$\sigma(t) = \frac{1}{1+e^{-t}}$$

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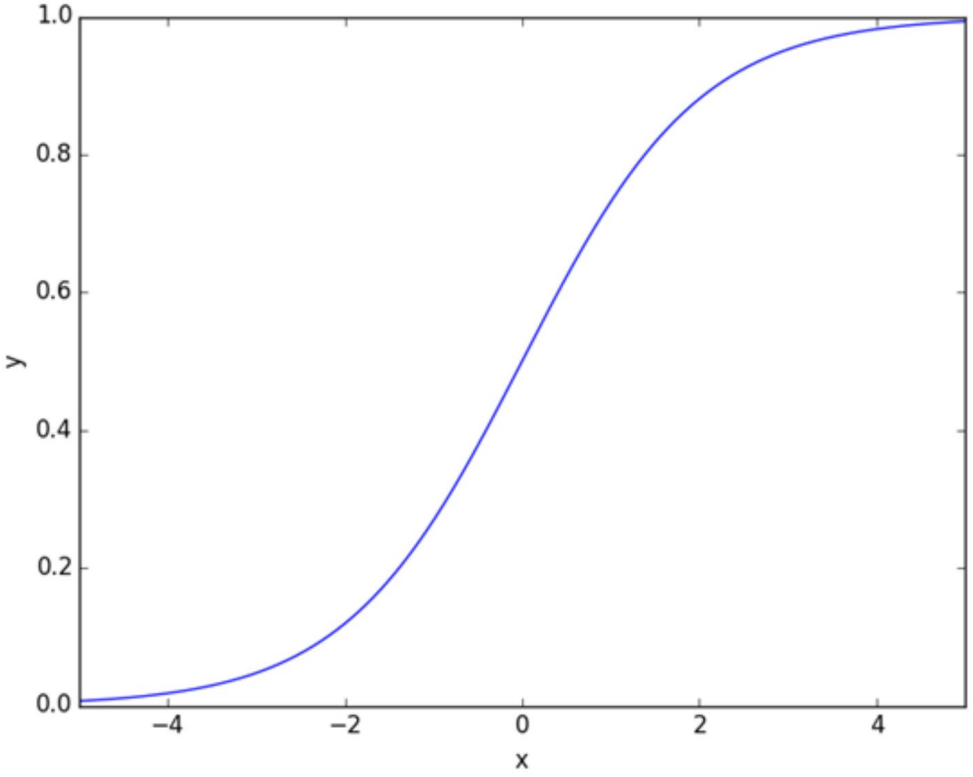
LOGISTIC REGRESSION FUNCTIONS

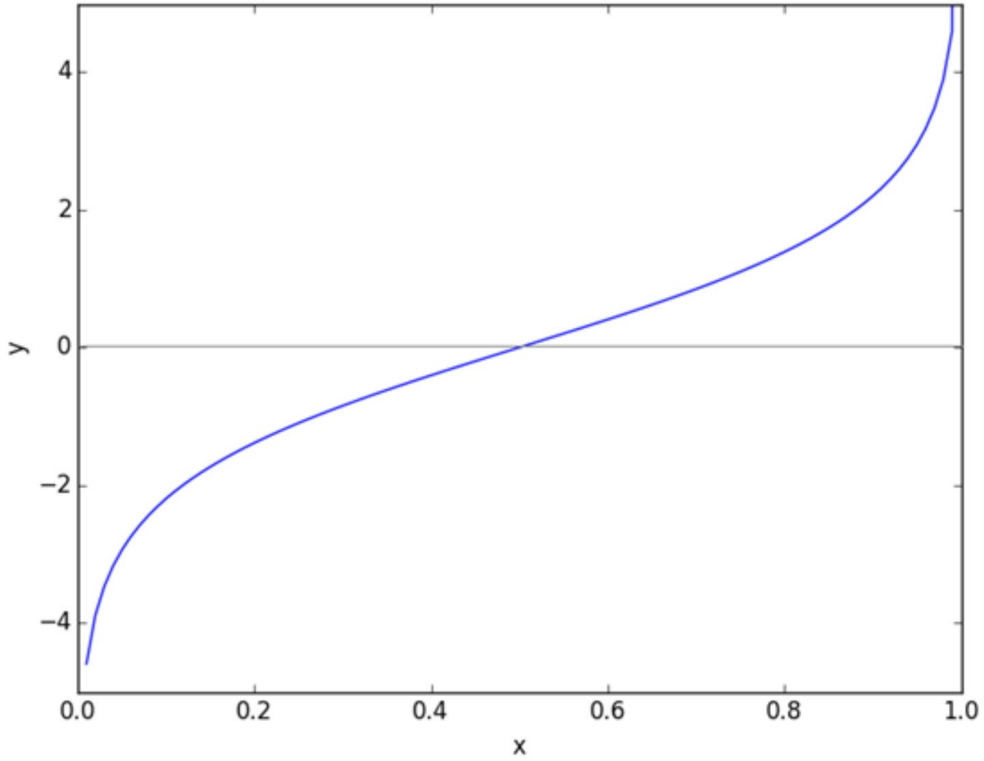
- **Sigmoid Function – Number to Probability**

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$



- **Logit Function – Probability to Number**

$$\text{Logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$$



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LOGISTIC REGRESSION

Problem Types:

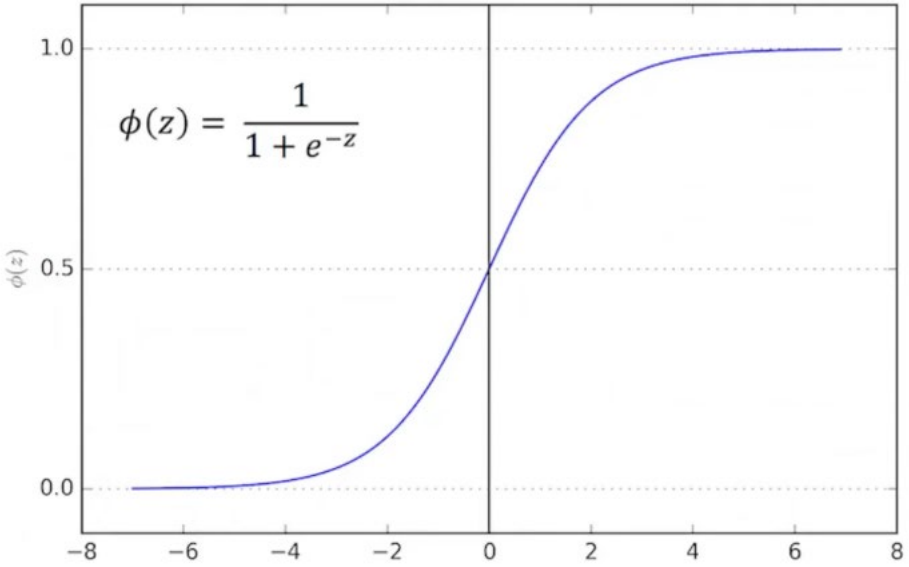
- **Binary Classification**: Predicts one of two possible outcomes (e.g., spam vs. non-spam emails).
- **Multi-Class Classification**: Extended by using techniques like One-vs-Rest.

Advantages:

- **Quick**: Able to apply to very large datasets
- **Interpretability**: Weights provide insights into feature importance.
- **Probabilistic Outputs**: Predicts probabilities for confidence estimation.

Limitations:

- **Linearity**: Assumes a linear relationship between input features and log-odds of the outcome.
- **Outliers**: Sensitive to outliers; regularization helps mitigate this.
- **Multicollinearity**: Correlated features can reduce interpretability.



INTERPRETING LOGISTIC REGRESSION COEFFICIENTS

- Each coefficient (β_i) represents the **change in the log-odds** of the outcome for a one-unit increase in X_i, **holding other variables constant**.
- Exponentiating gives the **odds ratio**:
  - $e^{\beta_i} > 1$: increase odds of the event
  - $e^{\beta_i} < 1$: decrease odds of the event
- Example:
  - If $\beta_1 = 0.7$, then $e^{\beta_1} = e^{0.7} \approx 2.01$
  - A unit increase in X_1 doubles the odds of the outcome!

$$\text{Logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$$

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Support Vector Machine (SVM)

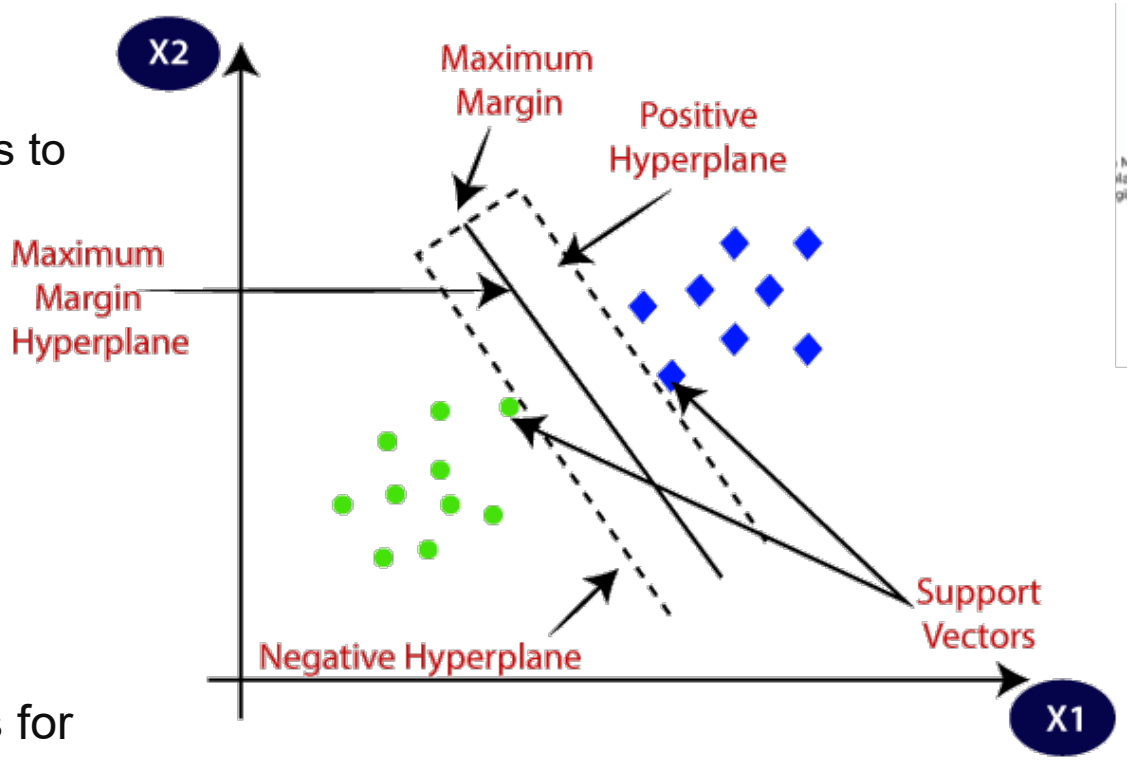
SUPPORT VECTOR MACHINE (SVM)

Core Idea: SVM finds the hyperplane that maximizes the (soft) margin between the closest data points (support vectors) of different classes.

- For non-linearly separable data, SVM uses kernel functions to project data into a higher-dimensional space where it becomes linearly separable.

Key Concepts:

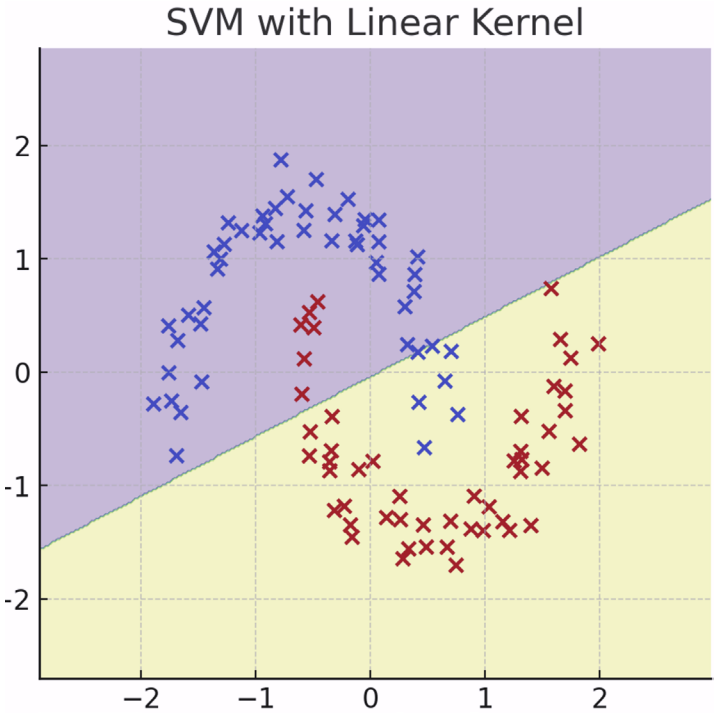
- **Hyperplane**: A decision boundary.
- **Margin**: Distance between the hyperplane and the nearest data points from either class.
- **Kernel Trick**: Transforms data into higher dimensions for linear separability. (This is often used for non-linear problems)



SVM Kernel Functions

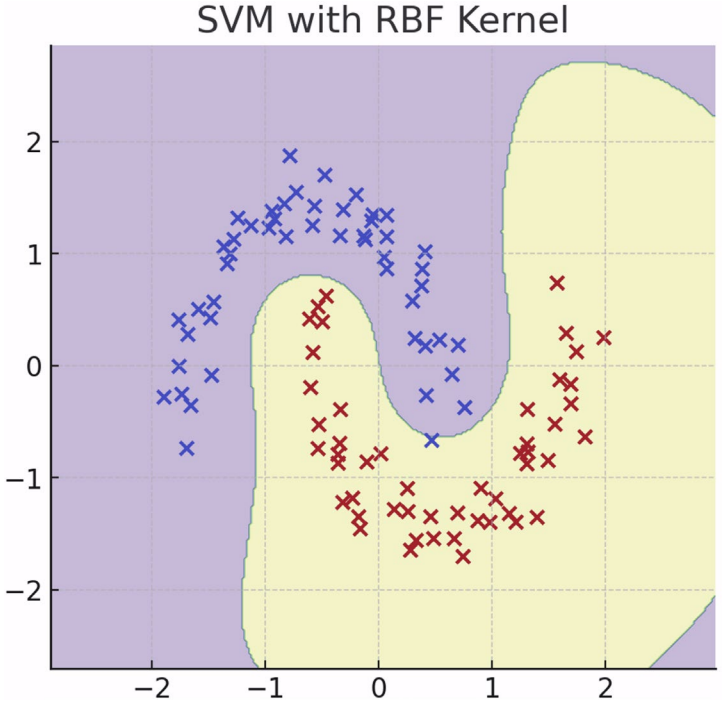
- Linear Kernel

$$K(x_i, x_j) = x_i^T x_j$$



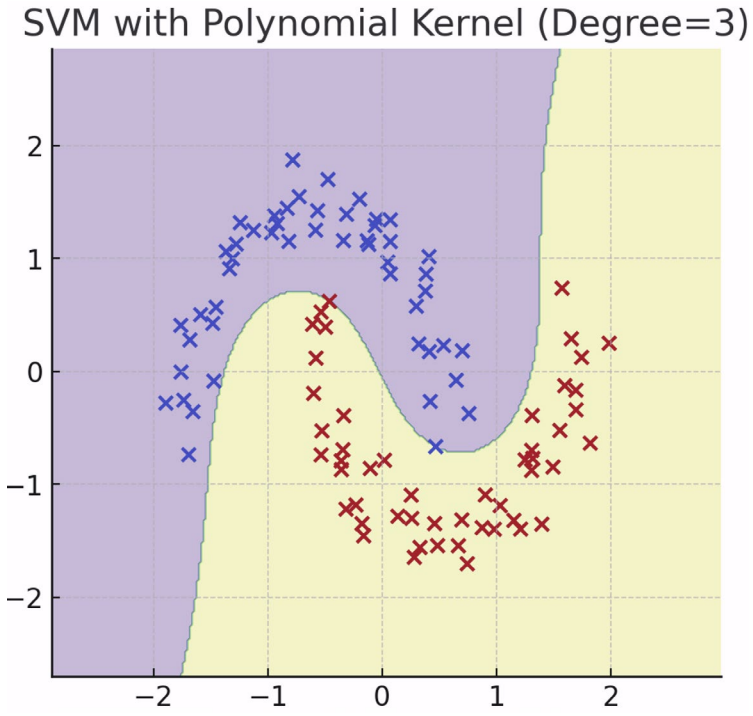
- Radial Basis Function (RBF)

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$



- Polynomial Kernel

$$K(x_i, x_j) = \left(x_i^T x_j + c\right)^d$$



SVM KEY PARAMETERS

- **C**: Regularization parameter.
  - High C means the model tries to classify all points correctly (narrow margin), while
  - Low C allows more misclassifications (wide margin).
  - **Opposite effects of choice of k w/ k-NN**
- **kernel**: Defines the type of hyperplane (linear, rbf, poly, etc.).
- **gamma**: Defines the influence of a single training point (applicable for RBF and polynomial kernels).
- **degree**: Degree of the polynomial kernel.

Advantages:

• Effective for high-dimensional data.

• Works well with a clear margin of separation.

Disadvantages:

• Computationally expensive for large datasets.

• Choice of kernel and hyperparameters can be highly problem dependent.

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Algorithm	Classification	Regression	Notes
Linear Regression	×	✔	Standard regression model; not suitable for classification.
Logistic Regression	✔	×	Despite its name, it's used only for classification.
Decision Trees	✔	✔	Versatile, handles both tasks by adapting the objective function.
Random Forest	✔	✔	Ensemble of decision trees; supports both tasks.
Support Vector Machines (SVM)	✔	✔	For regression, uses a variant called Support Vector Regression (SVR).
K-Nearest Neighbors (KNN)	✔	✔	Determines class (classification) or predicts value (regression) based on neighbors.
Naive Bayes	✔	×	Only suitable for classification due to probabilistic assumptions.
Gradient Boosting (e.g., XGBoost, LightGBM)	✔	✔	Boosting frameworks can handle both tasks efficiently.
Neural Networks (e.g., MLP)	✔	✔	Depending on the loss function and output layer configuration.
AdaBoost	✔	✔	Can perform both tasks but primarily used for classification.
Gaussian Processes	✔	✔	Provides probabilistic predictions for both tasks.
Linear Discriminant Analysis (LDA)	✔	×	Only designed for classification.
Ridge Regression	×	✔	A variant of linear regression with regularization.
Lasso Regression	×	✔	Regression method with feature selection through L1 regularization.
Polynomial Regression	×	✔	Extends linear regression for nonlinear relationships.
Bayesian Networks	✔	×	Primarily used for classification with probabilistic inference